Dir’a | دِرع

Sprint-0 Report

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1 Introduction

In recent years, online video games have evolved from mere entertainment to interactive digital environments in which millions of players worldwide communicate, collaborate, and compete directly across various platforms. These games are no longer solely individual activities but have become virtual social platforms reflecting real human interactions among people of different cultures and ages. While this interaction provides opportunities for learning and social bonding, it also opens the door to negative digital behaviors that can affect users' safety and psychological well-being, particularly among children and adolescents **[1].**

Video games enjoy widespread popularity both locally and globally, with children and young people making up a significant portion of users. For many families, age-rated games are considered a safe recreational option for their children. However, the reality within these games may differ from official ratings, as studies indicate that multiplayer game environments can involve behaviors such as cyberbullying, harassment, hate speech, and other harmful interactions [**2**]. Reports have shown that a noticeable proportion of players have experienced harassment or negative incidents while playing online **[3].** Since these ratings primarily evaluate pre-programmed content rather than unpredictable human behavior, this highlights a clear gap between theoretical age ratings and the actual experience of players within interactive environments.

The problem is not limited to the accuracy of age ratings; it extends to a more serious issue, where many games containing harmful interactive environments remain available for long periods without proper risk monitoring. Current monitoring mechanisms rely heavily on scattered individual reports or manual reviews, which delays the detection of dangerous behavioral patterns. Consequently, children may continue using games with offensive or harmful content without the knowledge of parents or relevant authorities. This fragmentation of data creates a significant gap between users and regulators, underscoring the urgent need for a unified platform that collects, analyzes, and clearly presents these reports to make the response faster and more effective.

Technology provides an effective opportunity to bridge this gap. By leveraging artificial intelligence and natural language processing, it is possible to analyze large volumes of user reviews and comments on different gaming platforms, such as app stores and distribution platforms, where users express their experiences, including exposure to bullying, toxic behaviors, or inappropriate content.

This type of data provides an indirect yet wide-ranging understanding of the interactive environment of each game. At the local level, the Kingdom of Saudi Arabia, within the framework of Vision 2030, seeks to enhance the role of video games and digital technologies in developing the digital economy and fostering local skills, while supporting innovation and creating a safe and attractive environment for Saudi youth in this sector **[4].**

This project proposes the development of a smart platform that measures and displays the risk level of each game based on the analysis of public reviews and comments, in addition to user reports within the platform itself. The system calculates a dynamic risk score for each game and identifies the detected risk categories, such as bullying, hate speech, inappropriate content. The platform also serves as a link between users and governmental authorities by aggregating player reports and clearly presenting high-risk games, helping relevant authorities respond more quickly and accurately. These transforms scattered individual reports into organized data that supports evidence-based decision-making.

The importance of this solution lies in its service to multiple stakeholders: it helps parents choose safer games for their children, enables regulatory authorities to monitor digital risks early, and contributes to improving the digital gaming environment, making it safer for young users. At the global level, this approach represents a shift from reactive monitoring, which relies on a large number of complaints before action, to a proactive method based on continuous data analysis. By combining community reports with intelligent content analysis, the platform helps build a safer and more transparent digital environment for future generations.

This document is structured to present the Dir'a project in a clear and systematic manner. Chapter 1 introduces the project by presenting identified problem, followed by the proposed solution and product overview. It includes the product vision, roadmap, objectives, and project scope. This chapter also describes the hardware and software tools used, the Scrum team structure, required skill sets, system roles and responsibilities Chapter 2 provides the project background, offering contextual and technical foundations related to online gaming environments and digital safety. Chapter 3 presents the literature review and competitive product analysis, examining existing solutions and highlighting the gap that Dir'a aims to address. Chapter 4 describes the system in detail, including user requirements elicitation, overall system architecture, and the use case diagram illustrating system interactions. Chapter 5 outlines the product backlog, including the structured backlog table

and defined non-functional requirements. Chapter 6 lists the references that support the research and development of the project.

1.1 The Problem

In the rapid expansion of digital gaming, games environments have evolved beyond visual gameplay into highly interactive social spaces. This evolution has exposed children to significant risks through player-to-player communication and unregulated user-generated content. Current monitoring mechanisms are largely reactive, relying on accumulated reports before action is taken. This delay allows harmful content to persist and potentially cause psychological harm. Research shows that timely moderation significantly reduces toxic behavior; for example, after Reddit banned hate-based communities, hate speech usage decreased by at least 80%. This highlights the importance of prompt intervention in reducing online harm. **[15]**

The scale of this issue is reflected in global and local statistics. According to a study by the Anti-Defamation League (ADL), about 62% of young gamers (ages 13-17) reported experiencing harassment while playing online multiplayer games **[5]**. Locally, in Saudi Arabia, approximately 47% of children reported experiencing or witnessing online bullying within video games **[6].** These figures highlight that nearly half of the child population in gaming environments is directly exposed to harmful interactions that current filters fail to catch in real-time.
This project specifically tackles the Reactive Monitoring Gap : a monitoring limitation in which oversight relies on delayed reactions, such as user complaints, instead of proactive complaints. Instead of waiting for a complaint to be filed after the harm has occurred, our project focuses on using Artificial Intelligence (AI) and Natural Language Processing (NLP) to proactively analyze player interactions and comments.

The platform specifically targets the protection of children and adolescents, with a primary focus on 7 years and above age group, who are statistically the most active and exposed to social risks in multiplayer environments**.** We aim to transform child protection from a manual, slow process into an automated, dynamic system that detects emerging risks (like extremist content or cyberbullying) immediately, providing early warnings to both parents and government authorities.

1.2 The Solution

To address the limitations of reactive monitoring and the lack of real-time visibility in digital gaming environments, this project proposes an AI-driven intelligent web platform that analyzes game-related data collected from multiple online sources. Rather than relying on manual user complaints or direct in-game access, the system aggregates information from publicly available platforms such as game reviews, player comments, online discussions, and curated datasets.

Using Artificial Intelligence (AI) and Natural Language Processing (NLP), the platform processes large volumes of text-based data to detect patterns related to cyberbullying, hate speech, extremist ideologies, violent discussions, and other harmful content. By applying trained classification models, the system assigns risk levels to each game.

The platform provides clear dashboards, alerts, and risk summaries tailored to different users:

- Government authorities access real-time overviews, detailed risk breakdowns, and reports management tools.
- Individual users can follow games, view risk for each game, compare risks, and receive early warnings about emerging threats.
- System administrators manage platform operations, monitor AI performance, manage user accounts, authenticate government employees, and ensure data integrity

By transforming scattered online information into structured, real-time intelligence, the platform closes the “reactive monitoring gap” and enables earlier detection of harmful trends before they escalate into serious psychological or social harm.

1.3 Product

1.3.1 Product Vision

For an individual user concerned about safety in digital games.

Who needs a reliable way to detect and understand risks such as cyberbullying, violent content, and harmful ideologies in online gaming environments.

The Dir'a is a web platform.

That analyzes publicly available game-related data to provide risk assessment, and age-appropriate recommendations.

Unlike traditional platforms such as Bark [14].

Our product delivers AI-driven risk intelligence across gaming ecosystems by analyzing data from multiple internet sources.

1.3.2 Product Roadmap

Figure 1-1:Roadmap

1.3.3 Objectives

This section outlines the objectives of the graduation project, categorized into product, project, and learning objectives.

A- Product Objectives (Customer-Focused Value)

1. **Provide AI-Based Game Risk Monitoring**
   Develop a platform that analyzes digital games and player comments using AI to detect risks such as harassment, violent language, and inappropriate communication in.
2. **Improve Awareness and Decision-Making**
   Enable parents and Individuals users to easily track games, view clear risk levels, age suitability, and receive alerts, helping them make safer gaming choices.
3. **Deliver Risk Alerts**
   Notify users when a tracked game's risk level changes, ensuring early awareness before harmful content escalates.
4. **Support Government and Regulatory Monitoring**
   Provide government employees with dashboards to monitor high-risk games, analyze trends, and manage reports to support faster and more informed regulatory action.
5. **Replace Static Ratings with Dynamic Risk Classification**
   Complement traditional age ratings (e.g., PEGI) with a dynamic, AI-driven risk score that updates based on real-time player-generated content.
6. **Enable Transparent Game Comparison**
   Allow users to compare games based on risk level, and category to identify safer alternatives.
7. **Promote Digital Safety Awareness**
   Increased awareness of hidden online gaming risks by presenting analytics, and risk explanations in a clear and accessible way.

B- Project Objectives (Solution-Focused Plan)

1. **Analyze User Needs and System Requirements**
   Identify requirements for parents, government employees, and administrators based on real-world digital safety challenges.
2. **Design a Multi-Role Platform Architecture**
   Implement role-based access control for parents, government employees, and administrators.
3. **Collect and Prepare Game-Related Data**
   Gather datasets such as game descriptions, and player comments, then clean and preprocess the data for analysis.
4. **Develop an AI-Based Risk Detection Model**
   Train and test an NLP-based model to classify content into different risk categories (e.g., low, medium, high risk).
5. **Build Interactive Dashboards and Interfaces**
   Design user-friendly dashboards for monitoring risks, managing reports.
6. **Implement an Alert System**
   Create a notification mechanism that sends alerts when new risks are detected in games.
7. **Test, Validate, and Deploy the System**
   Perform usability testing, AI accuracy evaluation, and system testing to ensure reliability and scalability.

C- Learning Objectives (Student-Focused Growth)

1. **Applied AI and NLP for Content Moderation**
   Gain practical experience in building and evaluating AI models for real-world safety applications.
2. **Full-Stack Platform Implementation**
   Strengthen skills in frontend design, backend development, database integration, and authentication systems.
3. **Multi-User Experience Design**
   Learn how to design interfaces tailored to different user roles with varying goals and permissions.

4. **Ethical and Responsible AI Development**
   Understand ethical considerations such as bias, privacy, and transparency when working with AI and sensitive user data.
5. **Integration of External Platforms and Standards**
   Gain experience working with external data sources and industry standards such as age rating systems.
6. **End-to-End Project Execution**
   Improve planning, documentation, testing, and presentation skills required for a complete graduation-level project.

1.3.4 Scope

This project aims to design and develop a web-based platform that leverages Artificial Intelligence (AI) to analyze textual content related to online PC games and assess their risk levels. The system will focus on user-generated content such as game reviews and comments to provide a data-driven evaluation of game safety for players aged 7 years and above.

The platform will be accessible through modern web browsers and will utilize widely adopted web and AI technologies, primarily Natural Language Processing (NLP) for text analysis. Direct integration with in-game chats or real-time voice communication is outside the current scope.

The high-level boundaries of the project include:

- **Technology and Platform:** Web-based application accessible via desktop and mobile browsers; no native mobile apps.
- **Language Support:** English textual content only; other languages may be considered in future versions.
- **Data Sources:** Publicly available game reviews and comments; private data or real-time in-game communications are excluded.
- **User Roles:** Parents/individual users, government employees, and system administrators with predefined access permissions.
- **Artificial Intelligence Capabilities:** Classification of textual content into risk levels (Low, Medium, High); no advanced behavioral analysis.

This scope ensures feasibility within the academic timeframe while delivering a functional solution that helps users identify and avoid risky PC games.

1.3.5 Hardware/Software Tools and Cost

Hardware Tools	
Name and Description	Cost
Personal Computers / Laptops	Already available – No additional cost (Available from project start)
Software Tools	
Name and Description	Cost
Web Development Tools (HTML, CSS, JavaScript) Backend Framework (Node.js)	Free
Backend Framework (Node.js)	Free
GPU / Compute Services – Used for AI model training and large-scale data processing.	190 – 1125 SAR
Paid APIs – Used to retrieve game data, reviews, and comments at scale.	75 – 375 SAR per month
Domain Name – Used to publish the Dir'a platform online.	50 – 120 SAR per year
Paid Dataset (Optional) – Used to enhance AI model training and improve classification accuracy.	200 – 750 SAR (one-time cost)
Generative Ai subscription (ChatGPT, Gemini , others...)	75 – 200 SAR per month
Miro platform for wireframes and outlines	40 SAR per month
Overleaf (docs editor)	367 SAR per year

Table 1:Cost

Other Relevant Costs

- **Cloud and Compute Costs:**
  Monthly cloud hosting and GPU usage represent ongoing operational expenses.
- **API Usage Costs:**
  Costs may increase depending on the volume of API requests and data retrieval.
- **AI Training Dataset Costs**
  A one-time cost may be incurred if premium datasets are used to improve AI accuracy.

  Note: The total project cost may exceed **SAR 1000**, mainly due to cloud hosting, GPU usage, and paid APIs, while hardware costs remain minimal.

Installed hardware & software:

The software required for this project includes installed development platforms used for writing and building the application, a server environment for running the system, a database management system for storing data, and the necessary libraries and frameworks that support AI implementation and text analysis.

1.4 Scrum Team

1.4.1 Skill Set Requirements

Technical Skill Required	What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan
Full-Stack Web Development	Intermediate
Database Management	Intermediate
API Integration	Intermediate, Studying the official documentation for each API to ensure smooth data collection of game reviews and comments.
AI & Natural Language Processing	Intermediate, Self-paced learning through online courses and practicing with the Kaggle dataset.
Data Analysis & Visualization	Intermediate, Learning how to use libraries to transform AI results into clear percentage-

	based breakdown charts.
Cloud Hosting & Deployment	Intermediate, Self-learning using official cloud documentation and practical tutorials to set up the server, database, and ensure 24/7 online access.

Table 2 : Skill set requirement

1.4.2 Learning

To enhance our technical capabilities and meet the project requirements, the team followed a structured learning approach aligned with the AI-based risk detection system. We evaluated our current skills in data analysis and identified areas that require further development. We will focus on deepening our understanding of API-based data collection mechanisms and risk classification techniques. We will also strengthen our knowledge of database design and management to ensure efficient and secure data handling. In addition, we have learned how to use Jira to organize tasks, track progress, and manage sprints professionally. We also plan to further develop new skills in Natural Language Processing (NLP) and Machine Learning Models to enhance the system's accuracy and improve its analytical capabilities.

1.4.3 Roles and Responsibilities

Scrum Team	
Product Owner:	Dr. Rana Almurshed
Developers:	Hatoon Alghamdi Nora Alkhudair Rathath Bin Rakan Malak Talal Basloom
Scrum Master (SM):	Dr. Rana Almurshed
Stakeholders:	Graduation Project Committee

Table 3:Roles and Responsibilities

2 Background

This project falls within the field of digital security and protection in online games environments, specifically in the domain of interactive digital games. In recent years, video games have evolved from simple individual entertainment tools into interactive social platforms that allow communication between players of different ages and cultures. This development has led to the emergence of new risks such as cyberbullying, harassment, hate speech, and harmful ideological content, particularly affecting children and adolescents.

The importance of this problem lies in the fact that children represent a large proportion of video game users and are often exposed to inappropriate content without the awareness of parents or regulatory authorities. Current systems rely mainly on manual reporting and delayed responses, which allows harmful content to persist for long periods before being detected and addressed.

Gaming Culture

Modern gaming culture refers to the contemporary environment in which video games function not only as entertainment products but also as interactive digital social spaces where millions of players communicate, collaborate, and compete in real time. Unlike early video games, which were mostly single-player and played offline, modern games are deeply connected to the internet and bring together players from different countries, cultures, and age groups. In this context, video games are digital applications designed for entertainment, competition, storytelling, or skill development. Today's games often include online multiplayer features, allowing many players to participate simultaneously in shared virtual worlds. These virtual worlds can be persistent, meaning they continue to exist and evolve even when individual players log off. As a result, gaming has become more than just a pastime.

Within this evolving landscape, modern gaming environments are often described as digital ecosystems. An ecosystem normally refers to a biological system where living organisms interact with each other and their environment. In gaming, a digital ecosystem similarly consists of interconnected elements such as players, game developers, communication tools, online communities, virtual economies, and user-generated content. All these components interact continuously, shaping the overall gaming experience. This interconnected structure highlights the technical foundations of gaming platforms.

Focusing on one of the main platforms, PC games (Personal Computer games) are video games designed to be played on personal computers, including desktop and laptop devices. They operate through computer operating systems and typically allow users to interact using a keyboard, mouse, or external controllers. PC games often provide advanced graphics, customizable settings, and online connectivity features, offering both single-player and multiplayer experiences. Such features further encourage player participation and interaction.

As online connectivity becomes an essential feature of modern PC games, players often interact with each other through dedicated gaming platforms and community services. Gaming platforms refer to digital systems that allow users to access games and communicate with other players over the internet. These platforms not only provide access to games but also support the creation of online communities where players can interact through features such as text and voice chat, reviews, and discussion forums. To maintain a healthy environment within these communities, gaming platforms typically implement policies and codes of conduct that regulate user behavior and help reduce abusive interactions.

As player participation increases, user-generated content (UGC) is any content created and shared by users rather than developers. It includes chat messages, reviews, comments, usernames, and other forms of content produced by players. Because players actively contribute content, regulation becomes more complex.

For this reason, traditional age rating systems such as PEGI or ESRB, which are based on fixed content before the game's release, this project uses a dynamic age classification based on risk level. This shift reflects the growing importance of live communication features.

Among these communication features, voice chat refers to real-time audio communication between players through microphones and headsets. It allows teammates to coordinate strategies, build relationships, and respond quickly during gameplay. In addition to voice communication, written interaction also plays a key role.

Similarly, text messaging or in-game chat allows players to send written messages to individuals or groups during or after gameplay. This supports communication across different languages and time zones. Beyond in-game interaction, players also connect on external platforms.

Expanding outside the game itself, online communities are groups of players who interact beyond the game itself through forums, social media platforms, streaming services, and fan groups. These communities help players share experiences, strategies, and content, strengthening social connections. Together, these elements form the broader gaming environment. While this complex ecosystem enables global connectivity, its open nature also gives rise to various potential risks.

In this broader context, the gaming environment refers to the overall digital space in which a video game operates. It includes gameplay mechanics, communication systems, player interactions, moderation systems, and all forms of user-generated content such as chat messages, usernames, and reviews. This environment reflects both the technical design of the game and the social behavior of its players. However, such complex environments also create potential risks.

One significant concern within these environments is player interaction, one of the most common social risks is cyberbullying, which involves repeated aggressive behavior carried out online with the intention of insulting, humiliating, or harming another player. This behavior may appear in the form of mockery, threats, offensive language, or continuous targeting of a specific player during gameplay or in chat. Closely related to cyberbullying is another harmful behavior.

In a similar way, another major concern is harassment, which includes any offensive or inappropriate behavior directed at another player. Harassment may involve threats, repeated insults, sexual comments, or discriminatory remarks based on gender, race, religion, or nationality. This type of behavior creates an unsafe and uncomfortable gaming environment, often causing victims to leave the game or avoid online interaction altogether. Beyond direct targeting, negative communication can also appear in broader forms.

More generally, toxic behavior is also widespread in many online gaming communities. It refers to hostile, disrespectful, or aggressive communication between players, even when it is not directed at a specific individual repeatedly. Examples include the use of offensive language, aggressive blaming of teammates, rage-filled messages, or disruptive behavior that ruins the experience for others. Although toxic behavior may appear less severe than bullying or harassment, it still contributes to a negative and unhealthy gaming environment. In addition to behavioral risks, content-related risks must also be considered.

From a content perspective, violent content represents another important risk. This includes scenes or gameplay elements that involve excessive violence, such as killing, torture, or intense combat

situations. While some level of action and conflict is common in many games, exposure to extreme violence especially when not suitable for the player's age may influence children's perception of aggression and reduce their sensitivity to violent behavior. Beyond violence, ideological influences may also emerge.

Another less visible but serious issue is ideological exposure, extremist or harmful ideologies represent a serious and less visible risk in online gaming environments. This includes any content, messages, or interactions that promote hate, racism, discrimination, or extremist beliefs. Such ideologies may appear in player conversations, usernames, symbols, or organized in-game groups. These examples demonstrate how risks extend beyond visible gameplay elements.

Psychological risks refer to the mental and emotional effects that gaming may have on players. These include addiction, anxiety, stress, reduced self-esteem, sleep disruption, and emotional harm caused by repeated exposure to hostile or negative communication.

Relevant Authorities in the Video Game sector

In Saudi Arabia, several authorities are involved in the video game sector and contribute to different aspects of the industry, including regulation, development, and support for gaming activities. The following section presents some of the key authorities related to this sector.

The Video Games Age Rating Department within the General Commission for Audiovisual Media, which operates under the Ministry of Media, is responsible for regulating and classifying video games in the Kingdom of Saudi Arabia. The department aims to protect society especially children and adolescents from inappropriate content by reviewing and evaluating video games before they are released in the Saudi market or on digital platforms. The system classifies games into several age categories such as 3+, 7+, 12+, 16+, 18+, and 21+, based on the nature of their content, which may include violence, inappropriate language, or sensitive themes. The department also ensures that publishers and retailers comply with these ratings and works to raise awareness among families about the importance of choosing age-appropriate games for their children. This system helps regulate the gaming market in Saudi Arabia while respecting the cultural and social values of the community.

Risk management

Before discussing risk management approaches and technologies, it is important to first understand the nature of risks within online gaming environments.

A risk is defined as any factor that may negatively affect a player's safety, psychological well-being, or overall experience. Because modern games are interactive and socially driven, risks are not limited to the game's visual content but also include behaviors and communication. These risks can be organized into broader categories.

Accordingly, in online gaming environments, risks can be categorized into several types, including social risks and content risks. Social risks arise from player-to-player interactions and include toxic behavior, harassment, cyberbullying, hate speech, and exposure to harmful or extremist ideologies. These risks are often difficult to control because they depend on human behavior rather than fixed game content.

In contrast to social risks, other risks relate directly to the game's material, which can be described as content risks**.** Content risks Are related to inappropriate or harmful elements within the game itself, such as excessive violence, sexual content, extremist themes, or disturbing narratives that may not be suitable for the intended age group. Given the diversity and complexity of these risk types, a structured approach is required to effectively manage and control them.

Accordingly, risk management becomes essential in online gaming environments. Risk management is the process of identifying, analyzing, evaluating, and controlling potential risks that may cause harm. In online gaming environments, risk management focuses on detecting and reducing harmful behaviors such as bullying, harassment, hate speech, and inappropriate content to ensure a safer digital space for users.

There are three main types of risk management approaches, each differing in the level of automation and human involvement.

The first type is Manual Risk Management, which relies entirely on human monitoring and user reports. In this approach, moderators review complaints and analyze content manually. While this method allows human judgment in decision-making, it is often slow and reactive, since action is usually taken only after harm has already occurred.

The second type is Semi-Automated Risk Management, which combines automated tools with human supervision. In this model, AI systems may flag suspicious or harmful content, but human moderators make the final decision. This approach improves response time compared to fully manual systems; however, it still partially depends on manual review and may not fully eliminate delays.

The third type is Automated Risk Management, which relies entirely on AI-based systems to automatically detect, analyze, and classify harmful content in real time without waiting for manual intervention. This approach is proactive and scalable, allowing faster detection of emerging risks and continuous monitoring of digital environments.

In Dir'a project, we apply Automated Risk Management, which relies on Artificial Intelligence (AI) to automatically analyze game reviews and detect harmful behaviors in real time. This approach enables continuous monitoring without depending on manual reports, allowing faster risk detection, dynamic risk scoring, and proactive alerts. Since risk management in online gaming depends on advanced intelligent technologies, AI serves as the foundation of our system by simulating human intelligence in analyzing, understanding, and classifying harmful content to ensure a safer gaming environment.

A major component of AI in this context is Natural Language Processing (NLP), which allows computers to understand, interpret, and analyze human language found in games reviews.

Supporting this process is Machine Learning (ML), a subfield of AI that enables systems to learn from large datasets and continuously improve their performance without being explicitly programmed for every scenario.

To achieve higher accuracy, many systems rely on transformer models, which are advanced deep learning language models, such as BERT, designed to understand context, meaning, and relationships between words in text with high precision.

To detect harmful content, systems rely on text classification, a machine learning process that automatically categorizes text into predefined classes such as bullying, harassment, violent language, or safe content. These classifications are part of risk detection models, which analyze patterns in games reviews and assign appropriate risk levels.

This process is known as risk classification, where content is categorized into levels such as low, medium, or high risk based on the type and severity of language used.

The analysis results are often summarized into a Risk Score, which represents the overall level of risk associated with a specific game based on the analyzed data.

To ensure timely responses, an alert system is used to send notifications to users, parents, or authorities whenever a significant risk is detected or when a game's risk level changes.

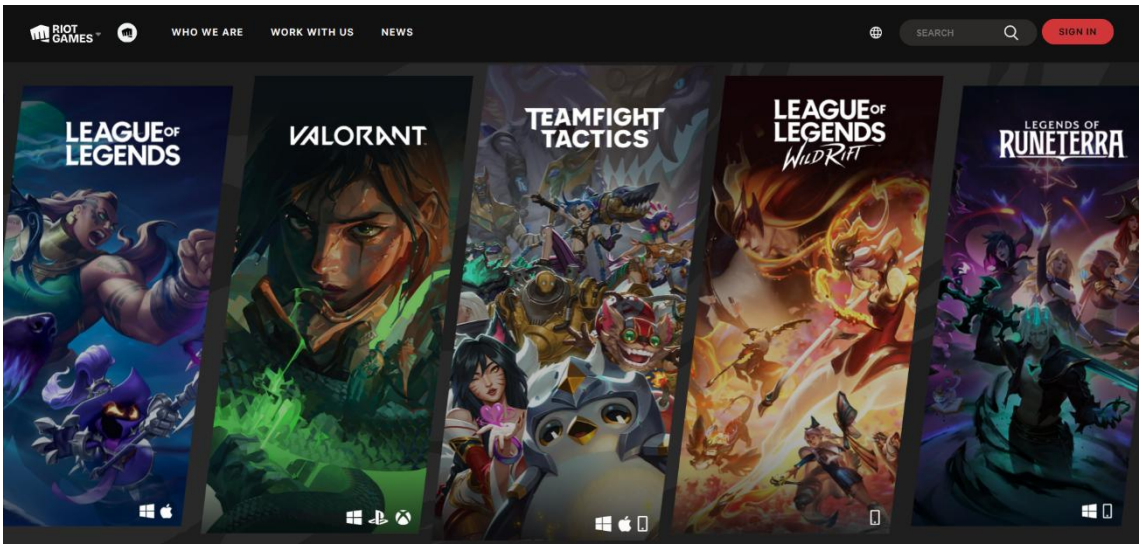
Dynamic age rating is a classification approach based on the continuous evaluation of risk levels within a game, where the appropriate age group is determined according to the analysis of player interactions and actual content within the gaming environment.

Classification Selection Methodology

The project does not rely on a single classification from a specific platform. Instead, it is based on an analysis of security policies and codes of conduct across several major digital platforms that manage gaming communities. By reviewing these policies, the most frequently observed patterns of abusive behaviors in in-game chats were identified and then grouped into a set of general categories representing the main forms of negative behavior in gaming communities.The following are examples of the platform policies that were analyzed:

1- Riot Games

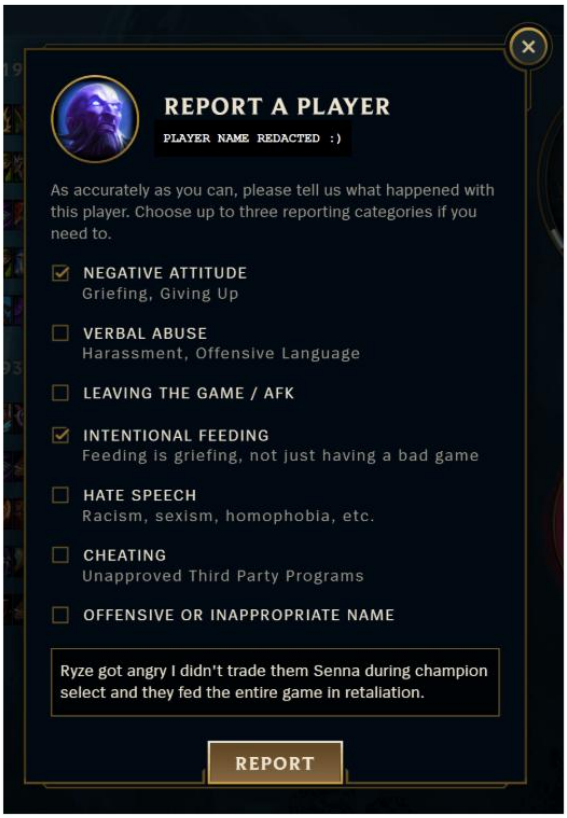
Definition: A global video game development and publishing company, Riot Games is well-known for competitive multiplayer games such as *League of Legends* and *Valorant*. The company employs advanced monitoring systems to track player behavior within its games and provides reporting tools and clear classifications for unacceptable in-game chat behaviors.



Policies: Riot Games defines a set of behaviors considered **reportable** within in-game chats. These behaviors include:

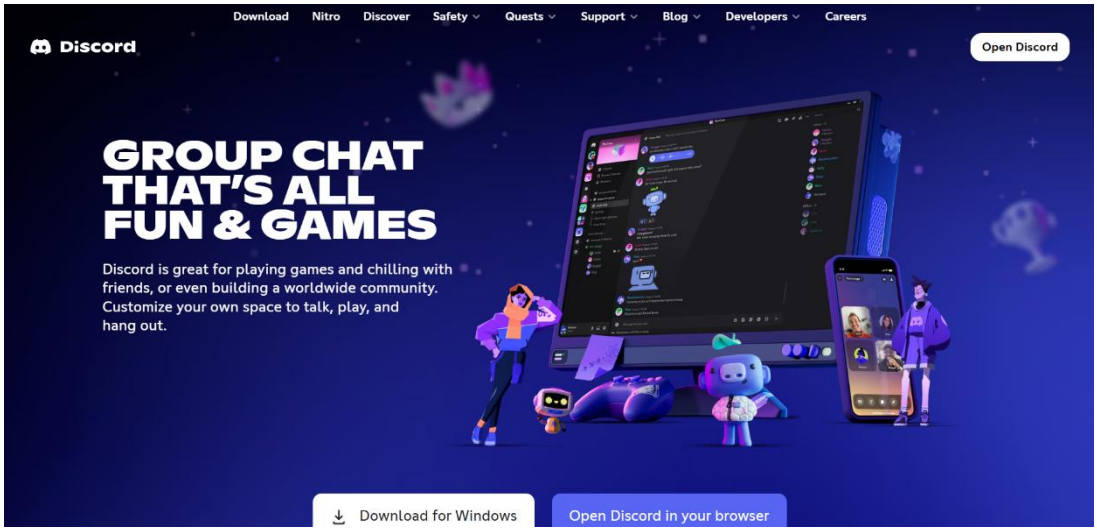
- Using insulting or offensive language directed at other players.
- Harassment or verbal abuse among players.
- Hate speech, such as racism or discrimination.
- Disruptive behavior that negatively affects the gameplay experience for others.
- Leaving a match or remaining inactive (AFK).
- Using inappropriate or offensive usernames.

These categories are reflected in the reporting interface of *League of Legends*, where players can report various types of misconduct.



2- Discord

Definition: A widely used social communication platform among gaming communities, Discord offers dedicated servers that allow users to communicate via text or voice. The platform also publishes a Safety Center that includes detailed policies on bullying, harassment, threats, and hate speech.



Policies: Discord's community policies outline a wide range of rules designed to maintain a safe and respectful communication environment among users. These policies cover many types of unacceptable behavior, including threats, hate speech, harassment, and violent or illegal content.

The platform emphasizes in its Respect Each Other guidelines that users must treat each other respectfully when interacting within servers or conversations. Some of the key behaviors prohibited by Discord include:

Harassment and Bullying

The platform prohibits any form of bullying or intentional harassment between users, including targeting individuals, attempting to intimidate them, or coordinating to harass them within servers.

Threats

The platform forbids threatening to harm any individual or group, whether the threat is direct, indirect, or even implied.

Hate Speech

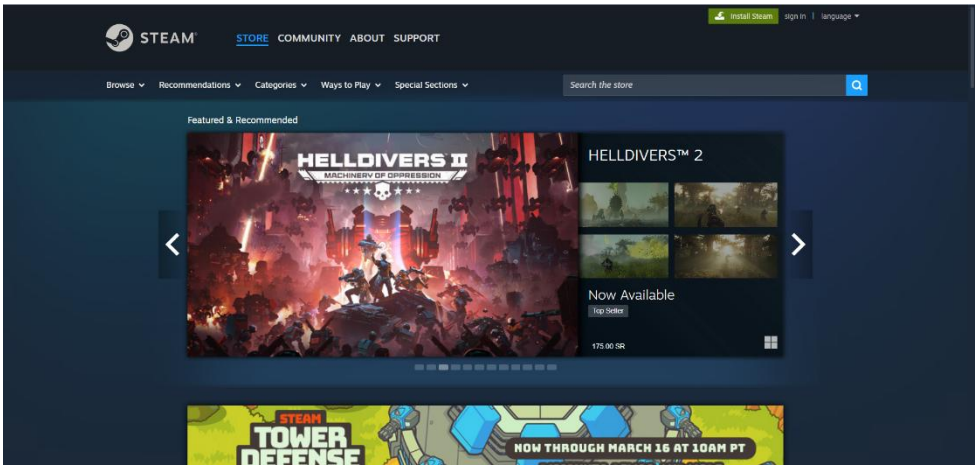
The platform prohibits the use of hate speech or any content that attacks individuals or groups based on protected characteristics such as race, religion, gender, or sexual orientation.

Doxxing and Personal Information Abuse

Sharing someone else's personal information without their consent, or threatening to publish it, is strictly prohibited.

3- **Steam:**

Definition: A digital platform for distributing PC games and managing gaming communities, Steam provides a variety of community features such as public discussions, user reviews, and user-generated content. The platform enforces codes of conduct to regulate interactions between players within these spaces.



Policies: Steam's community rules emphasize that interactions among players must be respectful and constructive. Unacceptable behaviors in discussions or reviews include:

- Insults or harassment toward other players.
- Threats or encouragement of harm.
- Discrimination or hostile content.
- Deliberate disruption or derailing of discussions.

These rules aim to maintain an appropriate community environment for all users within the global Steam community.

Based on the analysis of the above policies, the project adopts the following categories for analyzing in-game chat content:

- **Threats**: Phrases that contain threats of physical or psychological harm.
- **Hate Speech**: Abuse or discrimination against individuals or groups based on identity.
- **Sexual Harassment**: Unwanted sexually inappropriate or suggestive comments directed at a specific player.
- **Bullying**: Demeaning or insulting a specific player to humiliate them.
- **Toxicity**: Use of offensive or vulgar language in chat without targeting a specific player or identity.

These categories provide a general framework for analyzing negative behaviors in player interactions and are grounded in the common patterns observed across the security policies of major digital gaming platforms.

Theoretical Background

The project is based on several important theoretical concepts related to Artificial Intelligence and Natural Language Processing. These concepts form the foundation of the system that analyzes game reviews and detects harmful or risky content. The system uses AI and NLP techniques to analyze large volumes of textual data and classify risks such as bullying, hate speech, and violent discussions.

Among the core applications of AI in this context is text classification. Text classification systems use machine learning and deep learning algorithms to automatically assign text into predefined categories. In this project, text classification is applied to analyze game reviews and classify them into different risk categories, such as cyberbullying, offensive language, violent discussions, extremist content, or safe and low-risk content. The classification model learns patterns and linguistic features from labeled training data, then uses this knowledge to predict the appropriate category for new, unseen text. This enables the system to automatically detect harmful content at scale.

To enable accurate understanding of textual meaning, advanced language processing models are required. Language Models is an artificial intelligence model trained on large amounts of text data to understand, process, and analyze human language. These models learn grammar rules, word relationships, context, and patterns of both harmful and normal speech by analyzing large datasets. In this project, language models such as RoBERTa are used to understand the context of games reviews, detect harmful or risky language, and classify text into appropriate risk categories. Unlike traditional keyword-based systems, these models analyze the meaning of the entire sentence, which allows them to provide more accurate and context-aware predictions.

In addition to contextual understanding, analyzing emotional tone further strengthens risk assessment. Sentiment analysis models are NLP systems that determine the overall emotional tone of text by classifying it as positive, negative, or neutral. They learn linguistic patterns and contextual cues to understand whether a review expresses satisfaction, frustration, or anger. In Dir’a, sentiment analysis helps generate initial labels for game reviews and supports the risk scoring process, where strongly negative sentiment may indicate potential harmful behavior.

To specifically address abusive communication patterns, specialized detection models are employed. Toxicity and hate speech detection models are specialized NLP models designed to identify harmful language, including insults, threats, harassment, and hate speech. These models are trained on labeled datasets to recognize patterns associated with abusive behavior.

Beyond risk detection, the system also incorporates intelligent recommendation mechanisms. Recommender systems are algorithms designed to suggest items to users based on specific characteristics or criteria. In Dir'a platform, recommender systems are used to suggest safer alternative games and provide choices with lower risk scores. The recommendations are generated based on the calculated risk level of each game, its category, and the preferences or needs of the user. This helps parents and users make safer and more informed gaming decisions.

Finally, to ensure continuous data availability for analysis, external data sources are integrated into the system. In addition, external software services play an important role in data acquisition. The system can utilize Application Programming Interfaces (APIs) to retrieve data from relevant online gaming platforms and websites. These integrations enable the collection of reviews, user interactions, and publicly available content for analysis, supporting risk detection and classification processes.

3 Literature Review

3.1 Previous Studies & Research

1-Hate is No Game: Hate and Harassment in Online Games 2023 [3]

The study measures the prevalence of hate speech and harassment in online games, particularly through text and voice communication channels, and evaluates the effectiveness of content moderation systems implemented by gaming companies. It also examines the broader social impact of toxic experiences on players within these digital environments.

The findings reveal that a significant proportion of players experience harassment, with certain groups being disproportionately targeted. The study further identifies a gap between announced moderation policies and players' actual experience, especially in voice chat, highlighting the need for stronger monitoring systems and improved early detection mechanisms. These insights directly support the importance of structured risk assessment and effective oversight in online gaming environments.

This study directly supports the *Dir'a* project by identifying the level of risk in online games through measuring the prevalence of hate and harassment in gaming environments. The findings justify the need for content monitoring and evaluating the effectiveness of moderation systems, which represent core pillars of the *Dir'a* platform in assessing and displaying game risk levels.

2-Player versus Player: A Systematic Review of Cyberbullying in Multiplayer Online Games [7]

This study presents a comprehensive systematic review of cyberbullying in multiplayer online games, analyzing findings from 64 independent studies conducted across multiple countries and involving tens of thousands of participants, primarily adolescents and young adults. The review synthesizes evidence on the prevalence of cyberbullying and identifies psychological and behavioral characteristics associated with both perpetrators and victims. The findings indicate that cyberbullying is a relatively common phenomenon in competitive online gaming environments, and is associated with factors such as impulsivity, lower levels of empathy, and heightened competitiveness.

The study also emphasizes the role of anonymity in online gaming environments, suggesting that reduced accountability and perceived invisibility can increase aggressive and antisocial behaviors. Additionally, the competitive structure of Player-versus-Player (PvP) games may intensify hostile interactions, particularly in the absence of effective moderation systems. The authors propose a conceptual framework to guide researchers and developers in understanding and addressing cyberbullying in multiplayer gaming contexts.

This study supports the *Dir'a* project by examining the prevalence of cyberbullying in multiplayer online games and demonstrating that such behavior is relatively common among players. This highlights the necessity of a platform like *Dir'a* that communicates risk levels and raises awareness about potential exposure to harassment. Furthermore, the study identifies specific factors contributing to the spread of cyberbullying and notes that the absence of effective moderation systems in some games can intensify aggressive behaviors. This reinforces the need for a unified platform to monitor and analyze risk levels, helping game developers understand the root causes of rising risks and address them effectively.

3-Gamebullying Victimization and Depression Among MOBA Gamers [8]

This study collected empirical data from MOBA players (Multiplayer Online Battle Arena – competitive team-based online battle games) across different age groups, primarily adolescents and young adults, to examine their experiences with in-game bullying, including verbal harassment, exclusion, and targeted attacks. The research investigates the relationship between exposure to bullying and psychological outcomes, finding a significant positive correlation between frequent victimization and higher levels of depressive symptoms, as well as lower self-esteem.

The study also highlights the moderating role of psychological resilience, showing that players with higher resilience are less affected by the negative consequences of cyberbullying, but not completely immune to the harmful effects of sustained harassment. Overall, the research emphasizes the mental health risks associated with toxic social interactions in competitive online gaming and supports the importance of monitoring and intervention strategies to reduce these adverse psychological effects.

This study provides significant value to the *Dir'a* project from a psychological perspective by examining the impact of in-game harassment on players. It demonstrates that exposure to harassment increases levels of depression and lowers self-esteem. These findings justify the importance of risk assessment systems and the existence of a platform like *Dir'a* that displays the risk level of each game, enabling players to feel safer before selecting a game and potentially reducing exposure to psychological harm caused by cyberbullying.

4-Detecting Harassment in Online Games Using AI [9]

This study presents the design and implementation of machine learning models trained to distinguish between neutral and toxic verbal content in game communication channels. The models were evaluated using large datasets of annotated in-game speech, which enabled the identification of patterns associated with hate speech, insults, and aggressive expressions. The research shows that neural network-based classifiers and natural language processing (NLP) techniques can achieve high accuracy in detecting abusive communication in real time.

A key contribution of the study is the application of real-time analysis frameworks, allowing detection systems to operate effectively during live gameplay without significant latency. By integrating such systems with game platforms, moderators or players can be alerted as harassment occurs, enabling faster responses and reducing the negative impact on victims. This highlights the potential of AI-driven detection systems to monitor and flag toxic verbal interactions as they happen.

This study supports the *Dir'a* project from a technical standpoint by contributing to the development of tools for real-time harassment detection. It presents methods for using artificial intelligence and machine learning to identify toxic behaviors in text and voice chats. These approaches provide a technical foundation for designing an effective monitoring system capable of classifying risks and enabling early intervention before toxic interactions escalate.

5-Online Moderation in Competitive Action Games: How Intervention Affects Player Behaviors [10]

This study examines the effects of different forms of moderator actions, such as warnings, penalties, and the timing of interventions, on disruptive behaviors in games, including cheating, offensive usernames, and toxic communication via chat and voice. The findings indicate that moderation reduces disruptive conduct and discourages repeated negative actions, with quick and timely interventions being more effective than delayed ones. The severity of penalties also influences outcomes, with immediate penalties often providing stronger deterrent effects.

Overall, the research highlights that effective moderation systems can mitigate toxic behavior in competitive gaming environments and shape community norms by preventing prolonged participation of negative players. These results provide practical guidance for game developers and community managers to improve moderation practices and address toxic interactions more effectively.

This study benefits the *Dir'a* project by emphasizing the importance of moderation in reducing risks within online games. The findings show that immediate and rapid intervention significantly decreases aggressive and toxic behavior compared to delayed moderation. This supports the design of the *Dir'a* platform, which aims to display risk levels in real time and alert both players and game administrators when risk levels increase. Such an approach enables early intervention and more effective mitigation of toxic behaviors.

3.2 Competitive Product Analysis

To position Dir'a within the current digital safety landscape, a comparative analysis was conducted on seven existing solutions that address parental control, content filtering, or digital monitoring. These competitors were categorized into three primary groups based on their core functional focus:

1. **Content-Restricted Platforms (e.g., YouTube Kids, Netflix Kids)**
2. **Device-Level Parental Control Systems (e.g., Apple Screen Time, Google Family Link, Amazon Kids)**
3. **Social & Behavioral Monitoring Tools (e.g., Bark, Brandwatch)**

This classification highlights the structural differences between ecosystem-level gaming risk intelligence (Dir'a) and existing solutions that primarily focus on content restriction, device supervision, or social media monitoring.

1- Content-Restricted Platforms

Platforms that provide age-filtered and pre-selected digital content within their own ecosystem, focusing on limiting access to inappropriate material rather than monitoring user behavior.

1- Netflix Kids

Netflix Kids is a child-friendly version of the Netflix streaming platform designed to provide age-appropriate entertainment content. It allows parents to create dedicated children's profiles with restricted content based on maturity levels. Parents can also enable parental control settings such as PIN protection, viewing history monitoring, and profile restrictions.

As illustrated in **Figure 3-1**, Netflix enables profile separation through the "Who's watching?" interface, where parents can create and manage dedicated children's accounts. This profile-based structure allows content personalization and age filtering at the account level.

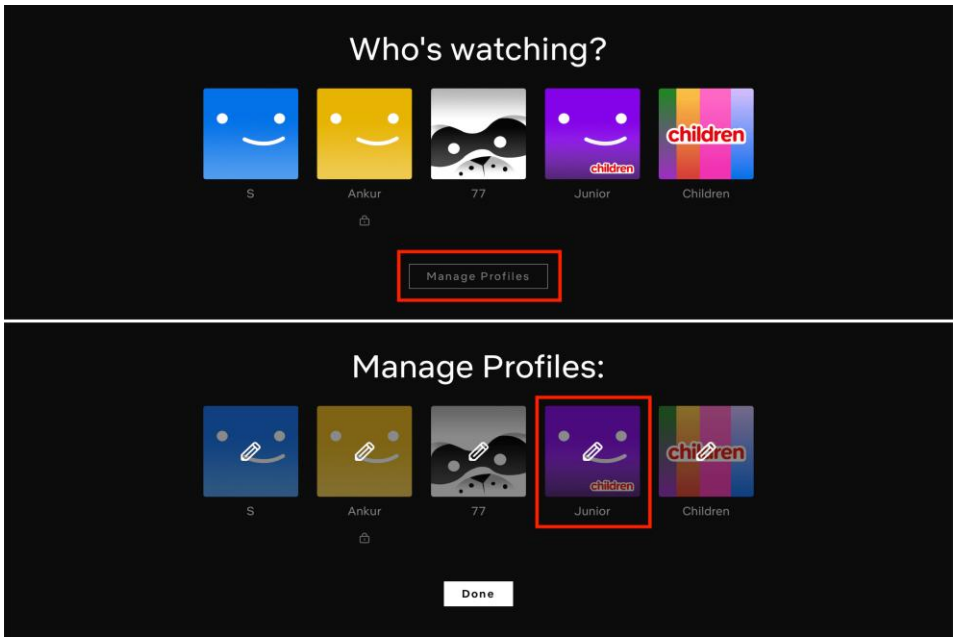


Figure 3-1: Netflix profile management interface

In addition, access to these parental control features is tied to the selected subscription plan. As shown in **Figure 3-2**, Netflix provides multiple pricing tiers (Basic, Standard, Premium), each offering different viewing capabilities and device limits.

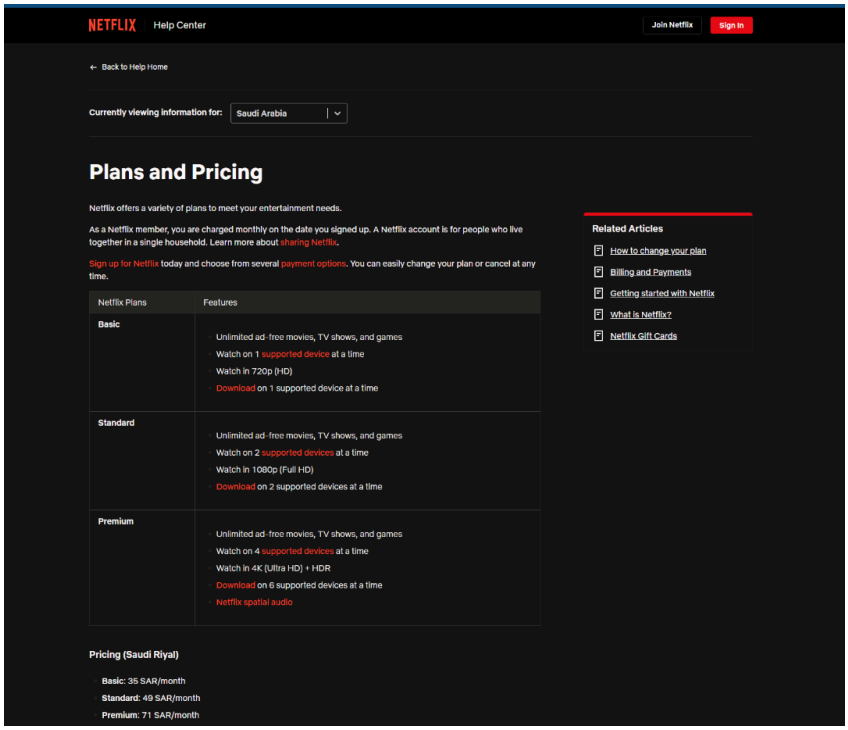


Figure 3-2: Netflix subscription plans and pricing

Key Features

- Dedicated children's profiles with maturity filters.
- Profile PIN protection.
- Content blocking based on age ratings.
- Viewing activity management.

Limitations

- Applies only within Netflix platform.
- Static maturity rating system.
- No AI-based risk classification.
- Does not evaluate multiplayer gaming risks.

2- YouTube Kids

YouTube Kids is a video-streaming platform designed specifically for children. It provides curated content and allows parents to configure filtering options based on age categories. Parents can choose content levels such as Preschool, Younger, or Older, depending on the child's age group.

As illustrated in **Figure 3-3**, parents can access control settings to select whether the child uses YouTube or YouTube Kids and define the appropriate content setting. This structured configuration allows age-based filtering within the YouTube ecosystem.

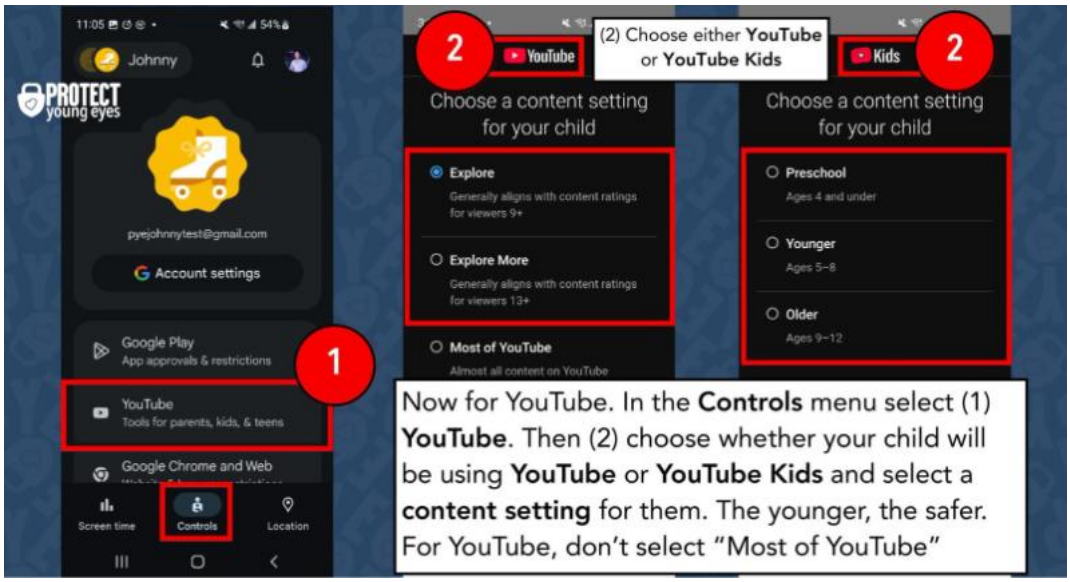


Figure 3-3: YouTube Kids content setting configuration showing age-based filtering options (Preschool, Younger, Older)

Furthermore, as shown in **Figure 3-4**, parents can manage additional general settings such as disabling autoplay, pausing watch history, pausing search history, or completely removing access to YouTube and YouTube Music. These controls focus on limiting content exposure rather than analyzing behavioral or gaming-related risks.

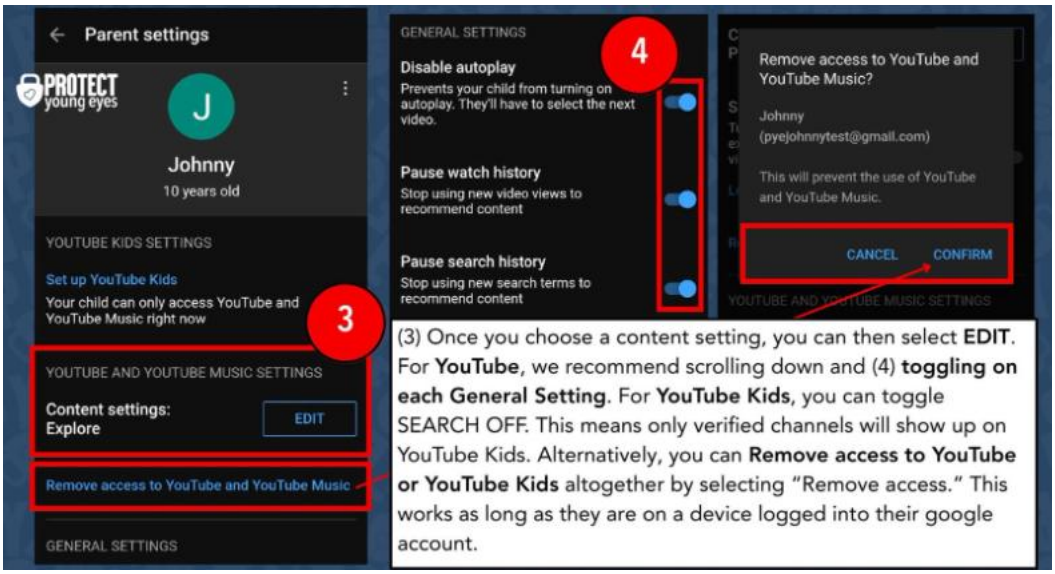


Figure 3-4: YouTube Kids parental general settings

Figure 3-5: YouTube Kids

Key Features

- Curated child-friendly video content.
- Age-based filtering categories.
- Search control and autoplay management.
- Screen time limits.

Limitations

- Limited to video content only.
- Does not analyze interactive gaming environments.
- No behavioral risk classification for games.
- No ecosystem-level AI risk scoring.

2- Device-Level Parental Control Systems

Tools that operate at the device level to manage screen time, restrict applications, and control content access through predefined supervision settings

1-Apple Parental Controls (Screen Time & Family Sharing)

Apple provides parental control features through **Screen Time** and **Family Sharing**, allowing parents to manage device usage, restrict applications, and monitor children's activity. These controls operate primarily at the device level and rely on predefined configuration rules rather than real-time AI-based analysis of gaming risks.

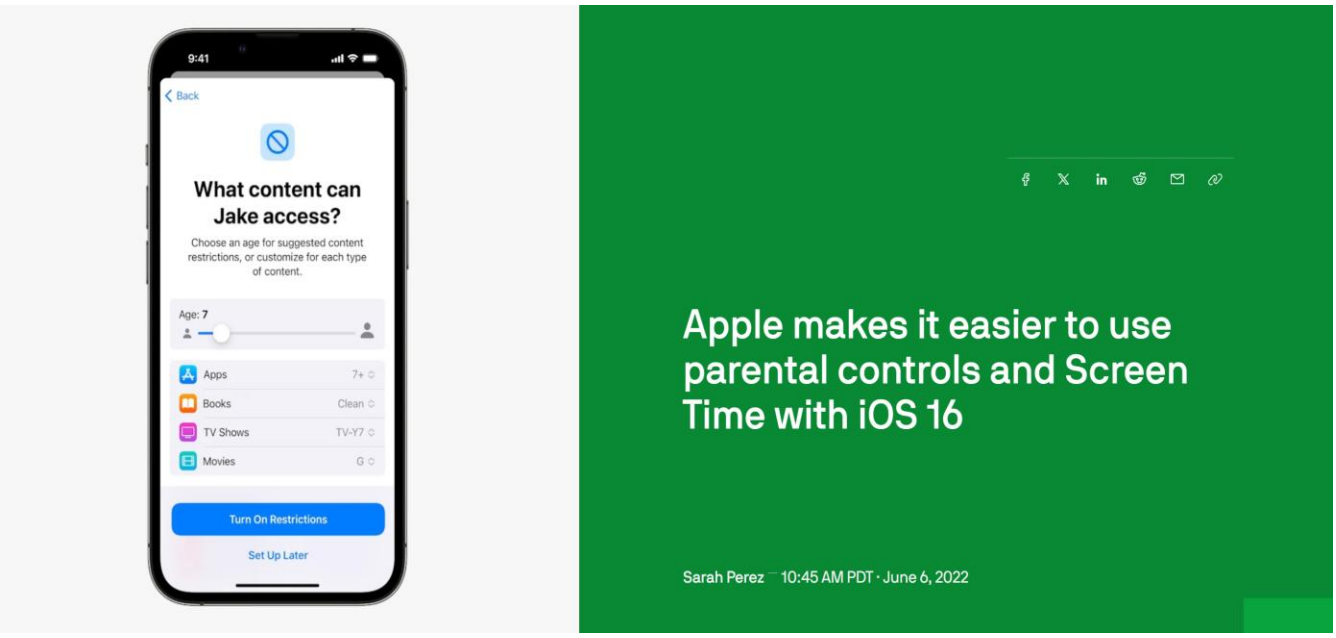


Figure 3-6: Apple Parental Controls

As illustrated in **Figure 3-7**, Apple allows parents to control access to multiplayer games, friend connections, private messaging, and profile privacy changes within the "App Store, Media, Web & Games" settings. This demonstrates Apple's focus on structured permission management.

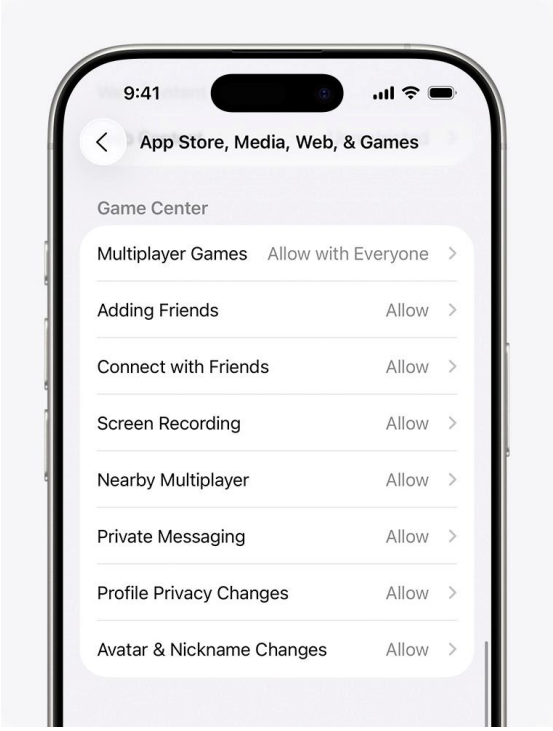


Figure 3-7: Apple Screen Time restrictions for games, multiplayer settings, and privacy controls.

To activate parental supervision, Screen Time must first be enabled. As shown in **Figure 3-8**, Apple provides options such as weekly activity reports, content & privacy restrictions, downtime scheduling, and app limits

Figure 3-8: Apple Screen Time activation interface showing restriction and reporting features

In addition, Family Sharing enables centralized management of children's accounts. As illustrated in **Figure 3-9**, parents can select a child's profile within the Family dashboard to manage permissions and device-level controls.

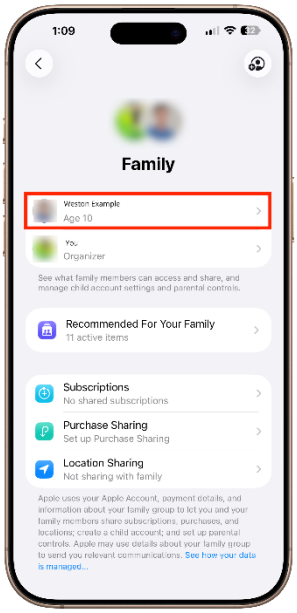


Figure 3-9: Apple Family Sharing dashboard for managing child accounts

Key Features

- App usage tracking and weekly activity reports.
- Content and privacy restrictions.
- App download and purchase approvals.
- Web content filtering.
- Game Center restrictions.

Limitations

- Limited to Apple devices only.
- Rule-based control without AI-driven behavioral analysis.
- No ecosystem-level gaming risk monitoring.
- No contextual or dynamic risk classification.

2-Samsung Family Link (Powered by Google)

Google Family Link is a parental control application developed by Google that enables parents to supervise their children's device activity. The platform allows parents to monitor app usage, approve or deny app download requests, set daily screen time limits, schedule bedtime restrictions, and track device activity.

As illustrated in **Figure 3-10**, Family Link provides a centralized parental dashboard that displays app activity reports, download approval requests, and scheduled screen time controls. This interface demonstrates Google's emphasis on structured device-level supervision and permission-based management.

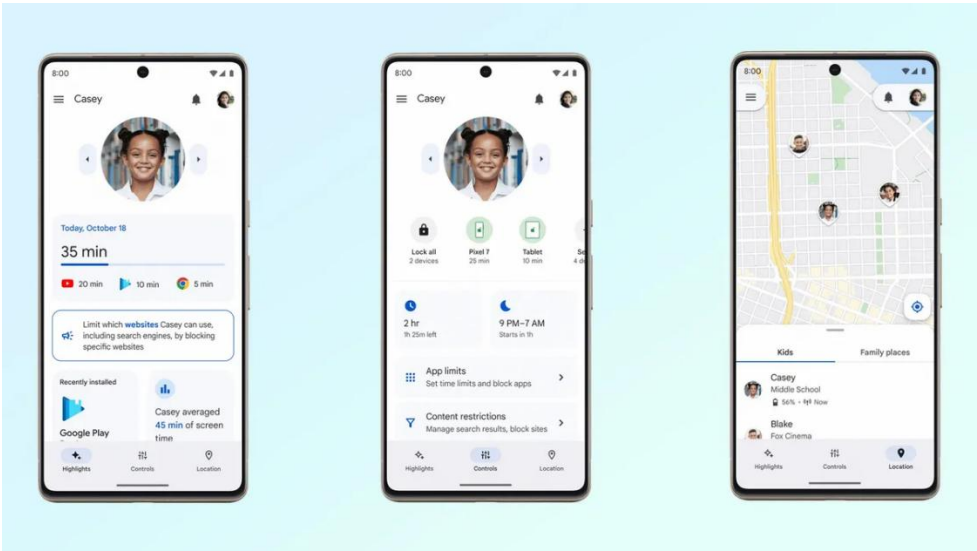


Figure 3-10: Google Family Link parental dashboard

Key Features

- Screen time limits and scheduling.
- App approval and blocking controls.
- Content filtering through Google services.
- Location tracking of child devices.
- Activity monitoring dashboard for parents.

Limitations

- Primarily device-level supervision.
- Focuses on app permissions rather than gaming behavior.
- No AI-based analysis of public gaming content.
- Does not provide risk classification for specific games.

3- Amazon Kids

Amazon Kids is a parental control service provided by Amazon that allows parents to manage and supervise children's digital activities across Amazon devices such as Fire tablets, Kindle, and Echo devices. It provides curated child-friendly content and enables parents to set usage limits and content restrictions.

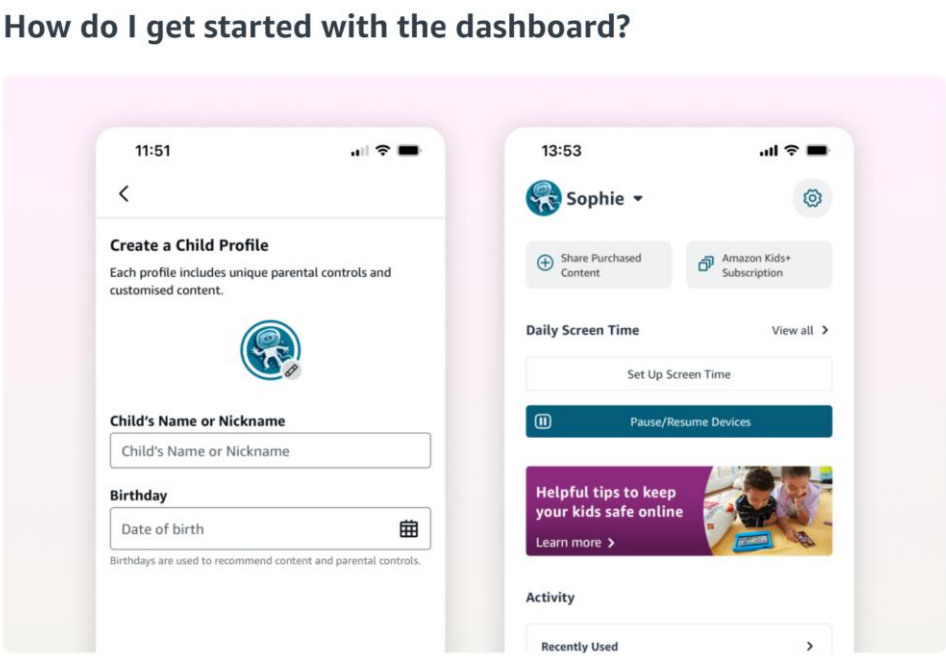


Figure 3-11: Amazon Kids dashboard

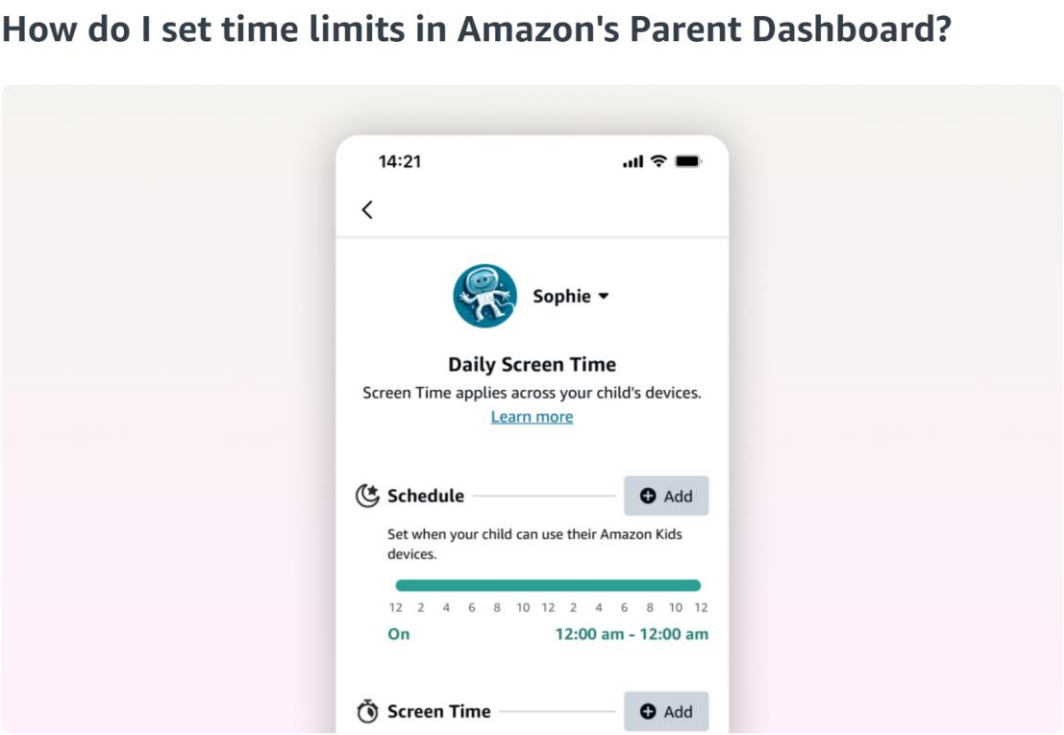


Figure 3-12:Amazon Kids time limits

Key Features

- Child profiles with age-based content filtering.
- Screen time limits and educational goals.
- Activity monitoring and content customization.
- Content restrictions across Amazon devices.
- Purchase approval controls.

Limitations

- Primarily limited to Amazon ecosystem devices and services.
- Focuses on content access rather than behavioral risk.
- No AI-based classification of gaming risks.
- Does not analyze public game reviews or ecosystem-level data.

3- Social & Behavioral Monitoring Tools

Solutions that analyze user communications and online discussions to identify patterns such as harmful interactions or sentiment trends, primarily at the account or platform level rather than across entire gaming ecosystems.

1- BARK

Bark is a digital parental monitoring service designed to help families supervise their children's online activity and protect them from potential digital risks. Instead of requiring parents to manually review every message, Bark uses Artificial Intelligence (AI) to automatically analyze text messages, images, emails, and social media interactions. When the system detects potentially harmful or concerning content, it sends an alert notification to the parent rather than displaying all communications.

The platform supports monitoring across more than 30 social media applications, in addition to SMS messages and email services, and operates across Android devices, iPhone, and computers.

As illustrated in **Figure 3-13**, Bark emphasizes centralized parental control by offering features such as content monitoring, real-time alerts, location tracking, screen time management, and website blocking.

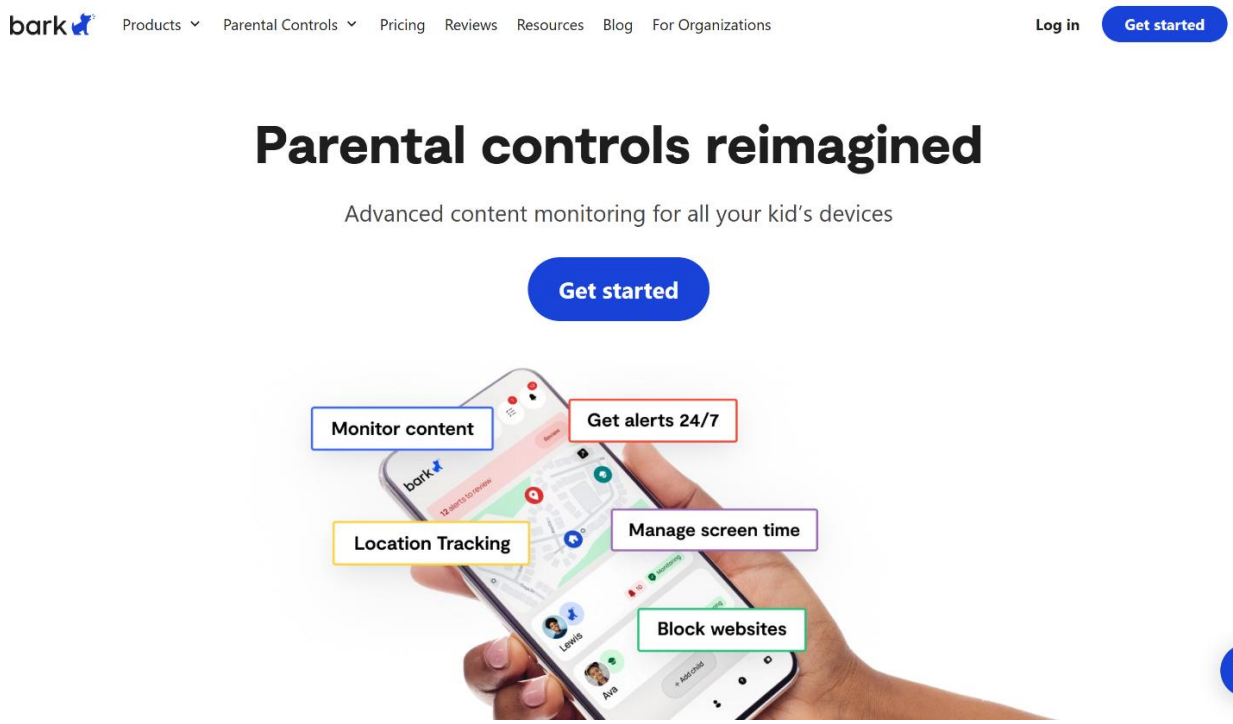


Figure 3-13: Bark parental monitoring interface

Key Features

- AI-powered monitoring of texts, emails, photos, videos, and social media activity.
- Real-time alerts when harmful or concerning content is detected.
- Monitoring across 30+ social media platforms.
- Screen time management and scheduling tools.
- Website and app blocking.
- Location tracking and alerts.

Limitations

- Primarily focused on device-level and account-level monitoring.
  - Reactive alert-based system rather than proactive game risk evaluation.
  - Does not provide dynamic risk scoring for games.
  - Does not analyze public gaming ecosystems.

2-Brandwatch

Brandwatch is a digital analytics platform primarily used by major corporations to understand public opinion about their brands. The platform collects and scans millions of online conversations across social media platforms, forums, blogs, and public review websites in order to determine whether a brand is being discussed positively or negatively.

Using large-scale sentiment analysis, Brandwatch enables organizations to measure brand perception, track emerging trends, and monitor competitors' performance within the market. Its core purpose is commercial intelligence and reputation management rather than child protection or digital safety.

As illustrated in **Figure 3-14**, Brandwatch provides centralized dashboards that visualize sentiment trends, audience insights, and social engagement metrics for marketing decision-making.

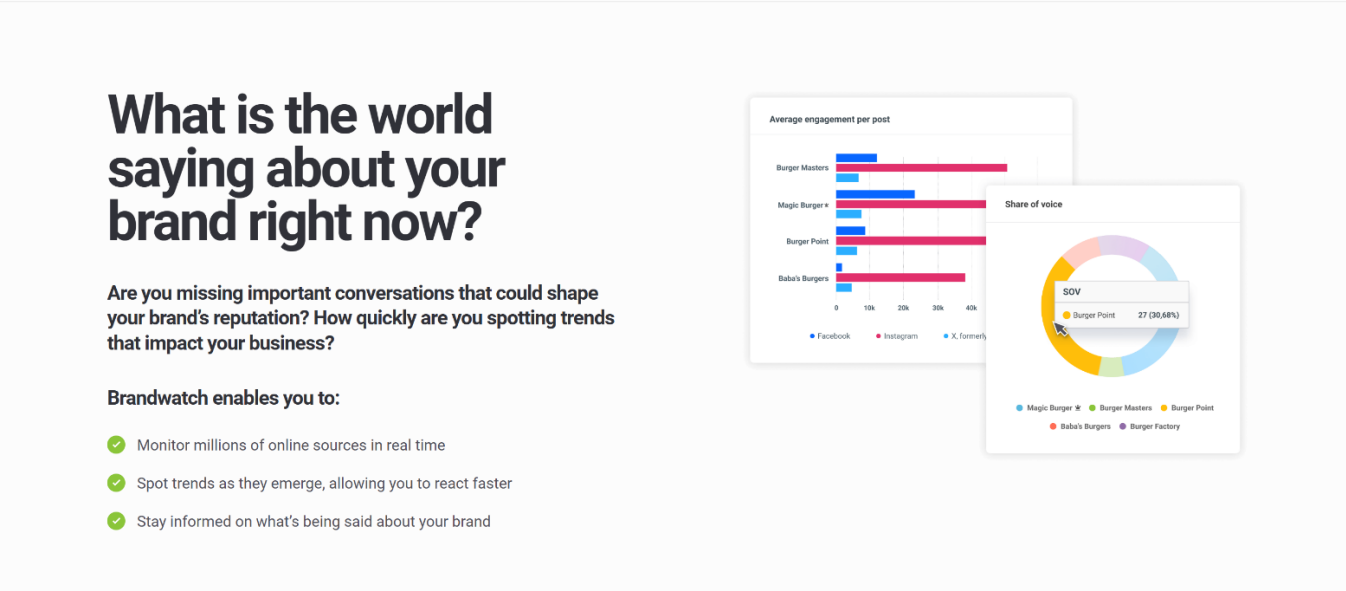


Figure 3-14: Brandwatch sentiment analysis dashboard

Key Features

- AI-powered social listening and media monitoring across social media platforms, forums, blogs, and review sites.
- Large-scale sentiment analysis to identify positive, negative, or neutral discussions.
- Trend detection and brand perception tracking.
- Real-time dashboards and analytics visualization tools.
- Audience segmentation and demographic insights.

Limitations

- Primarily designed for brand reputation management and marketing intelligence, not child protection or digital safety.
- Performs general sentiment analysis rather than structured risk classification.
- Does not categorize risks into specific gaming-related categories (e.g., bullying, hate speech).
- Not focused on PC gaming ecosystems specifically.
- Does not provide dynamic risk scoring for games.
- Does not support regulatory or government-level monitoring of gaming risks.

The following table compares existing solutions with Dir'a based on their core capabilities and system focus areas.

Features / Capabilities	Bark	Apple Screen Time	Google Family Link	YouTube Kids	Netflix Kids	Amazon Kids / Parental Controls	Brandwatch	Dir'a (Our Solution)
Device-Level Control	✓	✓	✓	✗	✗	✓	✗	✗
Screen Time Management	✓	✓	✓	✗	✗	✓	✗	✗
App Blocking / Restriction	✓	✓	✓	✗	✗	✓	✗	✗
Content Filtering	✓	✓	✓	✓	✓	✓	✗	✗
Messaging Monitoring	✓	✗	✗	✗	✗	✗	✗	✗
Social Media Monitoring	✓	✗	✗	✗	✗	✗	✓	✗
Gaming Ecosystem Monitoring	✗	✗	✗	✗	✗	✗	✗	✓
AI-Based Risk Classification	✗	✗	✗	✗	✗	✗	Sentiment Only	✓
Dynamic Risk Score	✗	✗	✗	✗	✗	✗	✗	✓
Risk Category Breakdown	✗	✗	✗	✗	✗	✗	✗	✓
Review-Based Risk Analysis	✗	✗	✗	✗	✗	✗	✗	✓
Pre-Play Risk Visibility	✗	✗	✗	✗	✗	✗	✗	✓
Government-Level Dashboard	✗	✗	✗	✗	✗	✗	✗	✓
Real-Time Risk Updates	✗	✗	✗	✗	✗	✗	✗	✓

Table 4 : Competitors' Features

4 System Description

In this section, we present the target users of the system and describe the methods used for gathering data. The study focuses on two main user groups: individual gamers and parents. Data was collected through interviews & questionnaires to understand user needs, behaviors, and concerns related to online gaming. The findings provide insights that guided the design and features of the system.

4.1 Users

The Dir'a platform is designed to serve three primary user roles, each with distinct characteristics, goals, and levels of technical expertise. Understanding these user profiles is fundamental to tailoring the system's functionality, interface, and access controls. The general characteristics of each user group are described below.

4.1.1 Parent / Individual User

This user group comprises parents, guardians, or any individual concerned about the safety of online gaming environments, particularly for children (aged 7 and above) and adolescents.

General Characteristics: This group is diverse and includes users with varying levels of technical proficiency. Their primary motivation is proactive players safety, not professional oversight. They are consumers of information who need clear, actionable insights to make informed decisions. They want a tool that provides simplified risk assessments They are typically busy individuals who value efficiency and clarity.

Educational Level: Varies widely, from elementary school graduates to post-graduate degree holders. The platform's design must therefore be intuitive and not require specialized technical or domain knowledge. They may have experience with game rating systems like PEGI or ESRB but are seeking a more dynamic and comprehensive safety metric.

Technical Expertise: Ranges from basic (browsing, using apps) to intermediate. The interface must be highly usable, with minimal steps required to find risk scores, track games, and understand alerts.

4.1.2 Government Employee

This user group consists of regulators, analysts, and decision-makers within governmental bodies responsible for digital safety, child protection, and media regulation (e.g., the General Commission for Audiovisual Media in Saudi Arabia).

General Characteristics: Users in this group are task-oriented and data-driven. Their goal is to monitor the digital landscape, identify high-risk games and emerging trends, and take action to report. They are accountable for the accuracy and timeliness of the information they use.

Educational Level: Typically hold university degrees, often in fields such as public policy, sociology, law, data analysis and related fields.

Experience: Professionals with strong awareness of child protection principles, experienced in assessing reported cases, objectively classifying risk levels, and making balanced decisions based on public safety standards. They demonstrate a clear understanding of harmful behavioral patterns such as harassment, exploitation, grooming, and toxic interactions in online environments.

Technical Expertise: Intermediate to advanced. They can navigate interfaces, and interpret data visualizations. They require access to raw data exports and detailed case management features, but the platform should present this in a structured, logical way to support their workflow.

4.1.3 System Administrator

This user group is responsible for the technical operation, maintenance, and integrity of the Dir'a platform. This role is internal to the project team.

General Characteristics: This user is technically focused, with responsibilities for user management, system configuration, data source integration, and monitoring the health of the AI models and the overall platform. They ensure the system is running correctly, securely, and efficiently. Their interaction with the platform is focused on backend functions rather than the core risk-assessment features.

Educational Level: University degree in Computer Science, or a related field.

Experience: Experienced in system administration, database management, and user access control. They have a strong understanding of web technologies and server environments.

Technical Expertise: Expert. They are comfortable with database queries. The administrative interface must provide them with the tools to manage users, monitor system performance, update data sources, and oversee the application's technical backend.

4.2 Requirements Elicitation

Dir'a's requirements were identified using both interviews and questionnaires to better understand the needs and concerns of gamers and parents. The interviews provided detailed insights into real gaming experiences and safety challenges, while the questionnaire collected broader feedback about user expectations and risk awareness. This approach helped ensure that the system is practical, user-focused, and aligned with real-world concerns. The following sections present the findings from both studies.

4.2.1 Interviews

We conducted interviews with five participants (Appendix A), including three teenage gamers and two parents. The purpose of these interviews was to better understand real gaming experiences, perceived risks in online environments, and expectations regarding an AI-based risk monitoring platform. The interviews were conducted in person and lasted between 15 to 30 minutes. A structured set of questions was prepared for each group (gamers and parents). Responses were manually documented and analyzed to extract recurring themes and insights related to toxic behavior, rating system limitations, and safety concerns.

Participants Demographic Information

Participant	Role	Age	Gender	Nationality
P1	Gamer	15	Male	Saudi
P2	Gamer	17	Male	Saudi
P3	Gamer	19	Male	Saudi
P4	Parent	47	Female	Saudi
P5	Parent	30	Male	Saudi

Table 5 : Participants demographic information

4.2.2 Interview Results

The interviews provided valuable insights into both player experiences and parental concerns.

From the gamers' perspective, toxic behavior such as insults, bullying, hate speech, and tribal discrimination was repeatedly mentioned as the most disturbing risk. Participants emphasized that current age rating systems focus mainly on violence and game content, but do not reflect real online community behavior. While some gamers rely on reviews, others prefer personal experience; however, most agreed that analyzing real player feedback would provide a more realistic safety assessment. They expressed interest in features such as clear risk levels, explanations of detected risks, violence level indicators.

From the parents' perspective, the main concerns included toxic language, exposure to strangers, cyberbullying, and the adoption of negative behaviors. Parents currently rely on age ratings, online reviews, supervision, and open-area gaming at home to monitor safety. However, they expressed

uncertainty about what happens during online interactions. Both parents indicated that a reliable AI-based platform should provide detailed explanations of detected harmful content, safer alternative recommendations, and guidance on contacting official authorities (e.g., cybercrime or child protection agencies) if necessary.

Overall, the findings highlight a clear gap between traditional age classification systems and the actual risks present in interactive online environments. The interviews strongly support the need for an AI-driven platform that continuously analyzes player-generated content and provides dynamic, transparent risk assessments to both gamers and parents. Full interview transcripts and detailed responses are provided in Appendix A (Interview 1,2,3,4,5).

4.2.3 Questionnaire

To understand the needs and concerns of both individual players and parents, an online questionnaire was conducted. A total of 35 responses were collected, of which 29 were PC gamers and 6 were parents (see Appendix 7-1). This structure was designed to capture a comprehensive view of the gaming environment. The gamer segment provided data on direct encounters with in-game risks and personal mitigation strategies, while the Parental segment focused on risk perception, and the necessity for advanced supervisory tools

Individual Users

Since this section focuses on individual users, the analysis is based on the 29 gamer responses. The detailed charts and statistical breakdown are provided in Appendix B.

Key Findings

The results revealed several significant insights:

- 82.8% of participants reported being concerned about behaviors inside online games (see Appendix 7-4).
- The most encountered issue was inappropriate language or toxic behavior (82.8%), followed by cyberbullying (55.2%) (see Appendix 7-5).
- 62.1% believe that traditional age ratings do not accurately reflect the real multiplayer environment (see Appendix 7-6).

- 65.5% expressed interest in knowing a game's risk level before playing it (see Appendix 7-7).
- When choosing a game, 69% prioritize user reviews, indicating that players value contextual explanations over static labels (see Appendix 7-10).

Additionally, many users reported that when encountering toxic behavior, they either report it or ignore it, highlighting the limitations of complaint-based moderation systems.

Conclusion

Based on the questionnaire findings, the project scope was refined to address the key concerns identified by individual users. Accordingly, the platform will:

- Move beyond static age rating systems by implementing a dynamic AI-based risk classification model.
- Analyze publicly available game reviews to detect harmful patterns such as toxic language, cyberbullying, and extremist content.
- Provide transparent explanations and categorized risk breakdowns, rather than relying solely on numerical labels.
- Support proactive decision-making by allowing users to view risk levels before engaging with a game.

These scope elements ensure that the system directly responds to user concerns while remaining aligned with the project's objective of addressing the reactive monitoring gap.
Full questionnaire questions and visual results are available in Appendix B.

Parents Users

This analysis is based on the 6 responses from parents whose children play PC games. The detailed charts and statistical breakdown of these findings are provided in Appendix B.

Key Findings

- The parental sample mainly represents guardians of children aged 7–11, a group characterized by very high gaming engagement (see Appendix 7-12). Notably, 83.3% of the children in this age range play PC games on a daily basis (see Appendix 7-13). This highlights how deeply gaming is embedded in their everyday routine.
- At the same time, anxiety levels among parents are extremely high. All parents in the sample (100%) expressed concern about behaviors occurring inside online games (see Appendix 7-

14). The most alarming issues identified were inappropriate language, extreme violence, and extremist content, each reported by 83.3% of respondents (see Appendix 7-15).

- A critical finding is the noticeable trust gap in existing rating systems. About 33.3% of parents explicitly stated that they do not believe official age ratings accurately reflect the real in-game environment (see Appendix 7-16). This skepticism directly contributes to the difficulty parents face when evaluating game safety. In fact, 33.3% reported struggling to trust official ratings, and another 33.3% highlighted challenges in identifying hidden features that are not immediately visible before gameplay (see Appendix 7-20).
- This uncertainty has created a strong demand for greater transparency. A large majority (83.3%) expressed the need to know a game's risk level before granting access (see Appendix 7-17). While age recommendations remain the top priority (100%), parents also place high value on detailed risk levels and user reviews (83.3%) as supporting indicators (see Appendix 7-19).
- In terms of technological solutions, openness toward AI-based platforms is relatively high, with 66.7% showing willingness to use such systems (see Appendix 7-18). More importantly, there was full consensus (100%) on the need for real-time alerts if a game's risk level increases after initial approval (see Appendix 7-21).

Conclusion:

The findings reflect a high level of parental anxiety regarding the potential risks their children encounter in the digital gaming landscape. This concern is primarily driven by a lack of adequate, transparent monitoring tools, which leads to significant difficulties for parents when trying to evaluate and choose appropriate games for their children. Consequently, there is a clear and strong endorsement for an AI-powered platform that can provide detailed risk assessments. By offering real-time alerts and specific breakdowns of content risks, such a platform would bridge the current safety gap. Ultimately, this would empower parents to overcome the challenges of game selection and make informed, confident decisions, ensuring a secure gaming environment for their children.

Summary for interviews & questionnaire:

The interviews and questionnaires helped us clearly understand the real concerns of gamers and parents regarding online gaming risks. Both groups expressed worries about toxic behavior, cyberbullying, inappropriate language, and the limitations of current age rating systems. The findings showed that traditional ratings do not fully reflect what actually happens inside multiplayer games.

Parents want more transparency, detailed risk explanations, and real-time alerts, while gamers prefer to know a game's risk level before playing it. These results confirm the need for a smarter and more proactive solution.

Therefore, the Dir'a platform aims to provide an AI-based system that analyzes game reviews, detects harmful patterns, and presents clear risk levels. This approach helps users make informed decisions and improves safety in online gaming environments.

4.3 Architecture

The system architecture directly affects the overall performance, scalability, and maintainability of the application. This section describes the architecture of our project, "Dir'a," which provides real-time game risk analysis and monitoring to support users and government authorities. The architecture defines how the system components interact to ensure efficient data processing and communication.

After evaluating several architectural approaches, we selected the Client–Server Architecture as the most suitable model for the Dir'a web platform. This model divides the system into two primary components: the Client Side and the Server Side, which communicate through HTTP requests to exchange data and results.

The primary system users include individual users (parents and gamers), government employees, and system administrators. Each role interacts with the platform according to predefined permissions.

Client Side

The Client Side represents the user interface accessed through web browsers by parents, individual users, government employees, and system administrators. It is built using HTML, CSS, and JavaScript to provide a responsive and interactive experience. Through this interface, users can search for games, view risk levels, compare safety ratings, receive alerts, and submit reports, while government employees access specialized dashboards for monitoring and analysis.

Server Side

The Server Side is responsible for the system's core logic and operations. It manages authentication, role-based access control, game data, and user reports, and communicates with the AI risk analysis

module. The server is built using a backend framework (Node.js) and is connected to a central database to efficiently process client requests.

API Integration

The platform integrates with external data sources through Application Programming Interfaces (APIs), such as the Steam API, which is used to retrieve game reviews and metadata in JSON format. The server sends requests to the API to collect up-to-date data, processes it, and forwards it to the AI risk analysis module. This integration ensures that the system relies on real-world, continuously updated data to improve evaluation accuracy.

AI Risk Analysis Module

The AI risk analysis module uses Natural Language Processing (NLP) to analyze user reviews and detect harmful content such as cyberbullying or hate speech. The content is classified into risk categories, and a dynamic risk score is calculated for each game. The results are then sent back to the server for storage and presentation.

Database Layer

The Database Layer stores all persistent data in a structured and secure manner, including user accounts and roles, game information, risk scores, user reports, and system logs. A relational database (MySQL) is used to ensure data integrity, fast retrieval, and scalability as the platform grows.

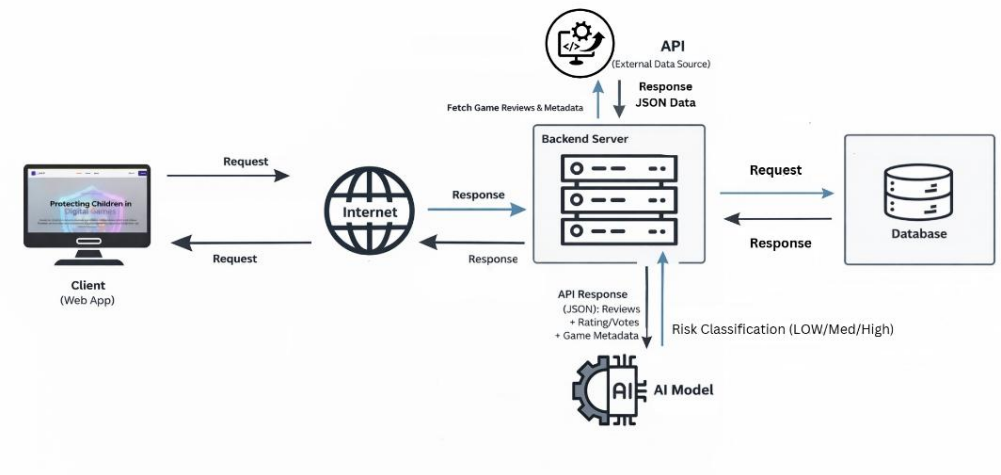


Figure 4-1:System Architecture Diagram

The architecture illustrates a system where a web application (Client) communicates with a backend server through structured request and response interactions. When a game is requested, the backend retrieves game reviews and metadata from an external API in JSON format. The data then undergoes preprocessing before being passed to the AI model for textual analysis and risk classification (Low/Medium/High) based on linguistic patterns and contextual meaning. A Risk Score is calculated and stored in the database for future reference and analytical purposes. Finally, the server returns the processed results to the client interface, where the risk level, review statistics, and supporting information are displayed within the dashboard. This design ensures a structured and automated workflow from data collection and analysis to storage and dynamic presentation, supporting informed decision-making.

Conclusion

The Client–Server architecture was selected because the Dir'a platform is developed as a web-based application, making this architecture the most suitable and efficient choice. It facilitates a clear separation of concerns by decoupling the user interface from AI processing and business logic, while enabling centralized risk analysis to ensure consistent and secure data handling on the server side. Additionally, this architecture provides the scalability required to support a growing number of users and games, and enhances security by protecting sensitive operations and authentication mechanisms from client-side manipulation. Therefore, after evaluating the project requirements, user roles, and system goals, the Client–Server architecture proves to be the most appropriate solution, offering a balanced combination of simplicity, scalability, security, and maintainability for delivering reliable AI-driven risk monitoring.

4.4 Use Case Diagram

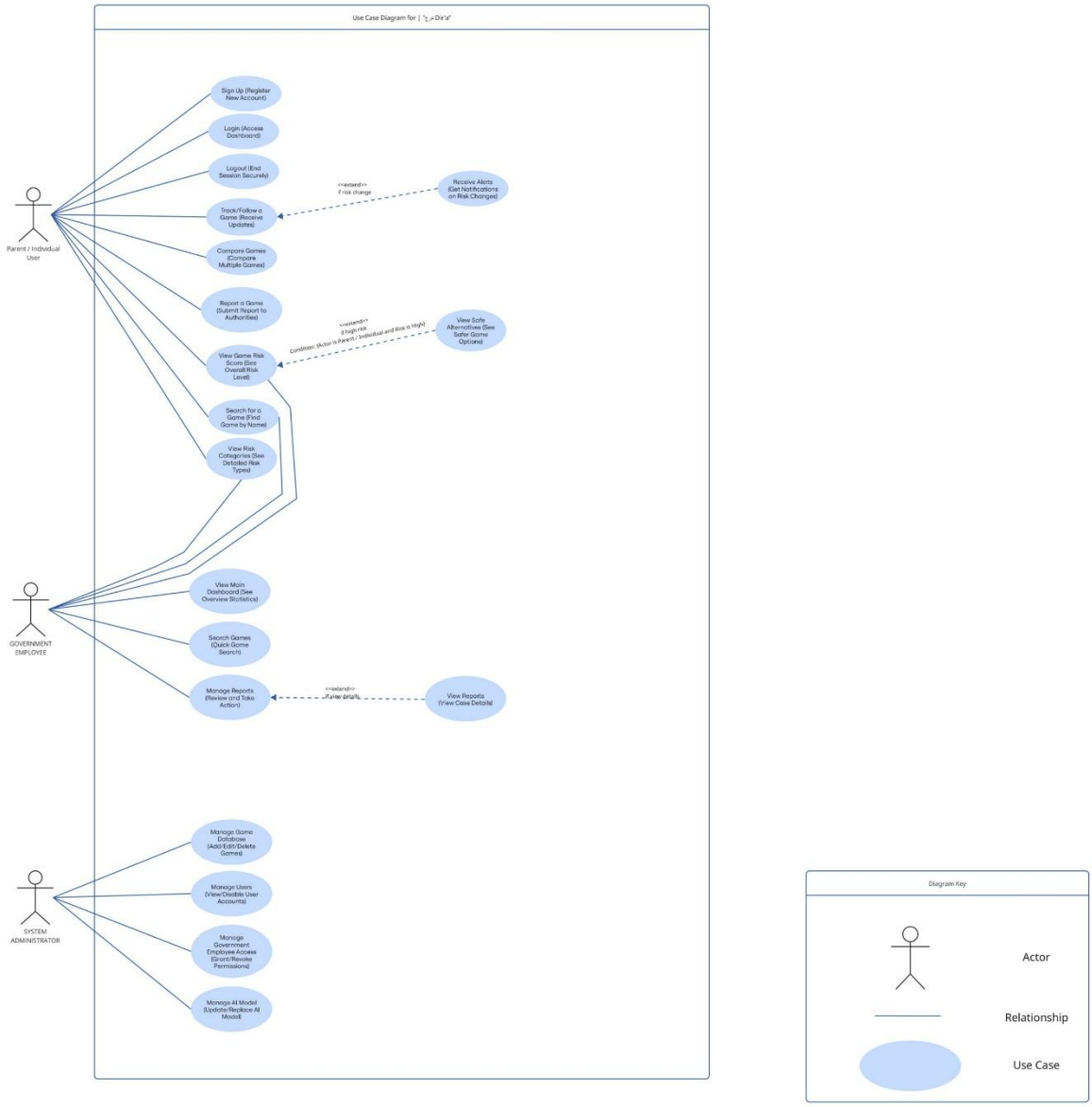


Figure 4-2: use case Diagram

5 Product Backlog

5.1 Functional Requirements

ID	PBI (User Story)	Size	Type	Acceptance Criteria	Status
1	As a user, I want to sign up so that I can access the platform features.	3	Feature	If I enter valid name, email, and password, then my account is created successfully. If the email already exists, an error message appears.	To do
2	As a user, I want to log in securely so that I can access my dashboard.	2	Feature	If credentials are correct, I am logged in. If incorrect, an error message appears.	To do
3	As a user, I want to log out so that I can end my session securely.	1	Feature	If I click logout, my session ends and I'm redirected to the home page.	To do
4	As a user, I want to search for a game so that I can check its risk level.	3	Feature	If I type a game name, matching results appear. If none are found, a "No results" message appears.	To do
5	As a user, I want to view a game's risk score so that I understand its safety level.	5	Feature	If I open a game page, the system displays the risk level (Low / Medium / High) and the overall risk percentage .	To do
6	As a user, I want to see the categories of risks detected so that I know the type of issues in the game.	5	Feature	If I view a game, I see categories like bullying, hate speech, violent content, etc.	To do
7	As a user, I want to compare two games I selected so that I can choose the safer option.	8	Feature	If I select two games, the system shows a comparison including risk level.	To do
8	As a user, I want to follow a game so that it is added	3	Feature	If I click "Follow," the game is added to my tracked list and notifications on	To do

	to my tracked list and I receive notifications .			by default.	
9	As a user, I want to unfollow a game so that it is removed from my tracked list and I don't get alerts.	2	Feature	If I click “Unfollow,” the game is removed from my tracked list and I don’t receive alerts.	To do
10	As a user, I want to report a game so that authorities can review harmful content.	5	Feature	If I click “Report,” a form appears. If I submit valid details, the report is stored successfully. If required fields are missing, an error appears.	To do
11	As a government employee, I want to view all risky games’ so that I can monitor dangerous trends.	5	Feature	If I access the government dashboard, I see a list of games sorted by highest risk level.	To do
12	As a government employee, I want to view detailed risk information so that I understand the dominant risk type.	8	Feature	If I select a game, I see percentage breakdown for each risk category.	To do
13	As a government employee, I want to manage reports about games so that I can take action.	8	Feature	If I review a report, I can see all report details , and i can mark it as reviewed or flagged.	To do
14	As an admin, I want automated AI analysis to run on game data so that risk scores are updated automatically.	5	Feature	If new reviews are added, the AI model automatically analyzes them and updates the risk score without manual intervention.	To do
15	As an admin, I want to	5	Feature	If I add a government employee with	To do

	manage government employees' access so that only verified government employees can access the system.			valid credentials, then the account is created with government access. If an unverified user tries to access the government dashboard, access is denied.	
16	As an admin, I want to view and edit registered users so that I can monitor and manage system usage.	5	Feature	If I access the user management page, I see a list of all users with their role and account status. If I edit a user's role or account status and click "Save," the changes are updated immediately in the system.	To do

Table 6 : Product Backlog for functional requirement

5.2 Non-Functional Requirements

ID	PBI (User Story)	Size	Type	Acceptance Criteria	Status
17	As a user, I want pages to load within 5 seconds so that the platform feels fast.	2	Feature	If I open any page, it loads within 5 seconds under normal conditions.	To do
18	As a user, I want the system available 99% of the time so that I can rely on it.	5	Feature	The system uptime is at least 99%.	To do
19	As a user, I want my data stored securely so that my account is protected.	3	Feature	Passwords are encrypted and secure authentication is enforced.	To do
20	As a user, I want the interface to be simple and intuitive so that I can use it without training.	1	Feature	A new user can search and view risk information within 10 minutes.	To do

21	As a user, I want the platform to handle at least 1000 concurrent users so that the system remains stable and responsive during peak usage.	8	Feature	System response time remains under 5 seconds when 1000 users are active.	To do
22	As a user, I want the platform to work on different browsers so that I can access it from any device.	3	Feature	If I open the platform using Chrome, Edge, or Safari, the system functions correctly without layout errors.	To do

Table 7: Product Backlog for non-functional requirement

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7 APPENCIES

7.1 APPENDIX A: Interviews

Gamers interviews:

Questions:

1. Could you briefly introduce yourself? (Age – Student/Employee – Your level of experience in PC gaming)
2. Approximately how many hours per week do you spend playing PC games?
3. What type of PC games do you prefer, and why?
4. Do you rely on other players' ratings and reviews? Why or why not?
5. Have you ever encountered toxic behavior or inappropriate content in a game? Please describe your experience.
6. What type of risk bothers you the most while gaming?
7. Do you think current rating systems truly reflect how safe a game is? Why or why not?
8. What do you think about the idea of a platform that uses AI to analyze game reviews and classify them based on risk level? What features would you expect it to have?

Answers:

Interview 1:

Interviewee: Ibrahim	**Interviewer: Nora**
Age: 15 **Nationality:** Saudi **Gender:** Male	**Appointment Date:** 20Feb **Start time**: 9:15 pm **End time:** 9:30 pm
Question:	**Answers:**
Q1. Could you briefly introduce yourself? (Age – Student/Employee – Your level of experience in PC gaming	I'm Ibrahim, 15 years old, a high secondary student. I've been playing PC games since I was around 10, so I would say I have good experience. I play a lot of multiplayer games and I understand how gaming communities work.
Q2. Approximately how many hours per week do you spend playing PC games?	Usually around 15–20 hours per week. On weekends or holidays it can be more, maybe 4–5

	hours a day.
Q3. What type of PC games do you prefer, and why?	I mostly prefer multiplayer games like FPS (first-person shooter), battle royale, and sometimes open-world games. I like competitive games because they're exciting and I can play with my friends.
Q4. Do you rely on other players' ratings and reviews? Why or why not?	Yes, sometimes I check reviews before downloading a new game. Reviews help me know if the game is fun or if it has problems like bugs or toxic players. But I don't fully trust ratings because sometimes people give bad reviews just because they're angry.
Q5. Have you ever encountered toxic behavior or inappropriate content in a game? Please describe your experience	Yes, many times. I've seen players insulting others, using bad language, and sometimes being racist. In voice chat, some players shout or say inappropriate things. Sometimes if you play badly, they blame you and start bullying. It can make the game less fun.
Q6. What type of risk bothers you the most while gaming?	The thing that bothers me most is toxic behavior and bullying. It's annoying when players are aggressive or disrespectful. Also, hate speech is really uncomfortable.
Q7. Do you think current rating systems truly reflect how safe a game is? Why or why not?	Not really. Age ratings usually focus on violence or content in the story, but they don't show how toxic the online community is. A game might be rated for teens, but the online chat can still be full of bad language and bullying.
Q8. What do you think about the idea of a platform that uses AI to analyze game reviews and classify them based on risk level? What features would you expect it to have?	I think it's a cool idea. It would help me know what's really happening in games before I play them. I would expect it to explain why the game is risky.

Table 8 : Interview 1

Interview 2:

Interviewee: Abdulrahman	**Interviewer: Hatoon**
Age: 17 **Nationality:** Saudi **Gender:** Male	**Appointment Date:** 22Feb **Start time** 7:30 pm **End time:** 8:00 pm
Question:	**Answers:**
Q1. Could you briefly introduce yourself? (Age – Student/Employee – Your level of experience in PC gaming	I am 19 years old and currently a student. I consider myself an intermediate-level gamer in terms of experience. I have good knowledge of different types of games and both multiplayer and single-player gaming experiences.
Q2. Approximately how many hours per week do you spend playing PC games?	I spend around 10 to 12 hours per week playing. During holidays, it may exceed 6 hours per day.
Q3. What type of PC games do you prefer, and why?	I prefer multiplayer games and story-based (single-player) games. Multiplayer games give me a sense of teamwork and excitement, especially in racing games. Story-based games provide an enjoyable and focused individual experience.
Q4. Do you rely on other players' ratings and reviews? Why or why not?	No, I do not rely much on other players' opinions because differences in perspectives are normal. I prefer to have my own personal experience and form my own opinion.
Q5. Have you ever encountered toxic behavior or inappropriate content in a game? Please describe your experience	Yes, while playing a multiplayer game, after defeating the opposing team, one of the players started insulting me using offensive language, which made me feel uncomfortable.
Q6. What type of risk bothers you the most while gaming?	What bothers me most is tribal discrimination among players, as well as account hacking or the use of fake accounts to harm others and steal their accounts or game progress.
Q7. Do you think current rating systems truly	No, I do not think so. Current systems mainly

reflect how safe a game is? Why or why not?	focus on age classification and do not fully reflect player behavior or safety within the game environment.
Q8. What do you think about the idea of a platform that uses AI to analyze game reviews and classify them based on risk level? What features would you expect it to have?	it is a very good idea. It should send notifications to parents when the risk level increases and allow them to submit official reports about specific games.

Table 9 : Interview 2

Interview 3:

Interviewee: Mohammed	**Interviewer: Hatoon**
Age: 19 **Nationality:** Saudi **Gender:** Male	**Appointment Date:** 21Feb **Start time** 8:30pm **End time:** 9:00 pm
Question:	**Answers:**
Q1. Could you briefly introduce yourself? (Age – Student/Employee – Your level of experience in PC gaming	My name is Abdulrahman. I am 16 years old and a student. I started playing PC games in 2019, and I consider myself an intermediate-level gamer. My favorite type of game is car racing games.
Q2. Approximately how many hours per week do you spend playing PC games?	I spend approximately 24 hours per week playing PC games, dedicating a significant portion of my free time. For me, gaming is not just entertainment; it is a way to enjoy challenges.
Q3. What type of PC games do you prefer, and why?	My favorite game is *Ark: Survival Evolved* because it is a non-realistic game that allows me to escape from daily life pressures. It offers an open world full of challenges and adventures, making the gaming experience enjoyable and intellectually stimulating, especially with its survival and exploration elements.
Q4. Do you rely on other players' ratings and reviews? Why or why not?	Yes, other players' ratings and reviews are important to me because they give me an initial

	idea about the quality and safety of the game. Reading others' experiences helps me understand the advantages and disadvantages before I start playing.
Q5. Have you ever encountered toxic behavior or inappropriate content in a game? Please describe your experience	I have not experienced a major issue directly; however, when joining online games with unknown players, there is always a possibility of being exposed to insults or bullying. This is frustrating because it negatively affects the enjoyment of the game and creates an uncomfortable environment.
Q6. What type of risk bothers you the most while gaming?	What bothers me most is tribal discrimination among players, as well as account hacking or the use of fake accounts to harm others and steal their accounts or game progress.
Q7. Do you think current rating systems truly reflect how safe a game is? Why or why not?	In my opinion, current rating systems provide an initial indication of a game's safety, particularly regarding age suitability or level of violence. However, they do not reflect the complete picture. They do not consider player behavior within the digital community or the actual experiences users may encounter during multiplayer gameplay.
Q8. What do you think about the idea of a platform that uses AI to analyze game reviews and classify them based on risk level? What features would you expect it to have?	I believe the idea of such a platform is excellent because it relies on real data instead of depending solely on official classifications. Analyzing thousands of real player reviews would provide a more realistic understanding of what actually happens within the game. Features I would expect from the platform such as Level of violence and Child safety level

*Table **10**: Interview 3*

Parents interviews:

Interview questions :

- Do you have children who play PC games? Approximately how many hours per day do they spend playing?
- How do you choose a new game for your children?
- How do you currently monitor the quality and type of content your children are exposed to while gaming?
- Since games have become interactive spaces, what concerns do you have about the risks your children might be exposed to?
- What kind of information would help you feel more confident about the safety of a game?
- If your child encounters aggressive behavior or inappropriate content, what actions do you usually take?
- Can you describe a specific situation where you felt unsure or concerned about a game your child was playing? What happened?
- What would you expect from a platform that uses AI to analyze gaming conversations and alert users to harmful content?

Answers :

Interview 4:

Interviewee: Tamader	**Interviewer: Nora**
Age: 47 **Nationality:** Saudi **Gender:** Female	**Appointment Date:** 21Feb **Start time**: 6:00 am **End time:** 6:30 am
Question:	**Answers:**
Q1. Do you have children who play PC games? Approximately how many hours per day do they spend playing?	Yes, I have two teenage sons who play PC games. On school days they play around 3–4 hours per day, and on weekends it can be 5 hours or sometimes more.
Q2. How do you choose a new game for your children?	I usually check the age rating first and read some online reviews. Sometimes I ask them to show me gameplay videos so I can understand

	what the game is about before allowing them to download it.
Q3. How do you currently monitor the quality and type of content your children are exposed to while gaming?	I try to talk to them regularly about what they are playing and who they are playing with. Occasionally, I sit with them while they play. I also check parental controls when available.
Q4. Since games have become interactive spaces, what concerns do you have about the risks your children might be exposed to?	I worry about toxic language, bullying, strangers talking to them, and exposure to inappropriate ideas or content. Online interaction makes it harder to control what they hear or see.
Q5. What kind of information would help you feel more confident about the safety of a game?	I would like clear information about the online community behavior, the level of toxic communication, and whether there are frequent reports of harassment or inappropriate content.
Q6. If your child encounters aggressive behavior or inappropriate content, what actions do you usually take?	I ask them to report the player and block them. If the problem continues, I may ask them to stop playing the game temporarily. I also discuss with them how to handle such situations calmly.
Q7. Can you describe a specific situation where you felt unsure or concerned about a game your child was playing? What happened?	Once, I overheard very aggressive language coming from the voice chat while my son was playing. I became concerned and asked him about it. He told me some players were insulting others. After that, we talked about muting voice chat and being careful online.
Q8. What would you expect from a platform that uses AI to analyze gaming conversations and alert users to harmful content?	I would expect it to provide detailed explanations of the type of harmful content detected.I would also like it to give recommendations for safer alternatives. Also it show which official authorities (like cybercrime or child protection agencies) can be contacted if needed.

Table 11 : Interview 4

Interview 5:

Interviewee: Hatem	**Interviewer: Hatoon**
Age: 30 **Nationality:** Saudi **Gender:** male	**Appointment Date:** 22Feb **Start time** 9:15 pm **End time:** 9:45 pm
Question:	**Answers:**
Q1. Do you have children who play PC games? Approximately how many hours per day do they spend playing?	Yes, I have a son who plays PC games. On regular days, he spends around 2-3 hours per day gaming. However, during weekends or holidays, the number of hours may increase depending on his schedule and academic responsibilities.
Q2. How do you choose a new game for your children?	I prefer to communicate with trusted people who have already played the game, such as relatives or close friends, and ask them about their experience in detail. I want to know whether the content and player behavior are appropriate and whether their experience was positive and safe before allowing my children to play it.
Q3. How do you currently monitor the quality and type of content your children are exposed to while gaming?	I do not allow my children to play in a closed room or separately from the family. Instead, I make sure they play in an open area of the house where all family members can see what they are playing. This allows me to monitor the type of content indirectly and ensure a safe and transparent gaming environment.
Q4. Since games have become interactive spaces, what concerns do you have about the risks your children might be exposed to?	I am concerned about account hacking or sharing personal information without awareness, as well as the possibility of adopting negative behaviors from multiplayer gaming environments.
Q5. What kind of information would help you feel more confident about the safety of a	The type of information that would make me feel more confident about a game's safety includes a

game?	clear and detailed evaluation that goes beyond age classification. I would like to know the level of violence, the nature of player interactions
Q6. If your child encounters aggressive behavior or inappropriate content, what actions do you usually take?	I the first thing I do is speak with him calmly to understand exactly what happened. I make sure to support him emotionally and ensure he was not negatively affected. Then, I block the offending player or use the in-game reporting system.
Q7. Can you describe a specific situation where you felt unsure or concerned about a game your child was playing? What happened?	es, there was a time when I noticed my son seemed tense and withdrawn after finishing a gaming session. When I asked him about it, he told me that some players had mocked him and used inappropriate language during an online match.
Q8. What would you expect from a platform that uses AI to analyze gaming conversations and alert users to harmful content?	The type of information that would make me feel more confident about a game's safety includes a clear and detailed evaluation that goes beyond age classification. as well as the availability of effective protection and reporting tools.

Table 12 : Interview 5

7.2 APPENDIX B: Questionnaires

Questionnaire questions:

1. **Which of the following describes you?**
   A. I play PC games
   B. I am a parent of a child who plays PC games

As Gamer:

2. **What is your age group?**
   A. 7–11
   B. 12–15
   C. 16–17
   D. 18–20
   E. 21+
3. **How often do you play PC games?**
   A. Daily
   B. Several times per week
   C. Rarely
4. **Have you ever felt concerned about behaviors inside online games?**
   A. Yes
   B. No
   C. Not sure
5. **Which types of risks have you encountered? (Select more than one)**
   A. Cyberbullying
   B. Inappropriate language or behavior
   C. Very violent content
   D. Extremist content
   E. I am not encountered

6. Do you believe age ratings reflect the actual in-game environment?

A. Yes
B. No
C. Not sure

7. Would you like to know a game's risk level before playing it?

A. Yes
B. No
C. Maybe

8. How do you usually deal with harmful or toxic behavior in online games?

A. Ignore it and continue playing
B. Mute or block the player
C. Report the behavior
D. Leave the game
E. I have never experienced this

9. Would you use a platform that shows AI-based risk levels for games?

A. Yes
B. No
C. Maybe

10. What information matters most to you when choosing a game?

A. Risk level score
~~A.~~B. Type of risks
~~B.~~C. User reviews
~~C.~~D. Age recommendation
~~D.~~E. Content explanations
~~E.~~F. How to play

11. Would you prefer to receive alerts when a game's risk level increases?

A. Yes
B. No
C. Not sure

As Parent:

1. **What is your child's age group?**
   A. 7–11
   B. 12–15
   C. 16–17
2. **How often do they play PC games?**
   A. Daily
   B. Several times a week
   C. Rarely
3. **Have you ever felt concerned about behaviours inside online games?**
   A. Yes
   B. No
   C. Not sure
4. **Which types of risks concern you the most? (Select more than one)**
   A. Cyberbullying
   B. Inappropriate language or behavior
   C. Very violent content
   D. Extremist content
   E. I am not concerned
5. **Do you believe age ratings reflect the actual in-game environment?**
   A. Yes
   B. No
   C. Not sure
6. **Would you like to know a game's risk level before allowing your child to play it?**
   A. Yes
   B. No
   C. Maybe

7. **Would you use a platform that shows AI-based risk levels for games?**
   A. Yes
   B. No
   C. Maybe

8. **What information matters most to you when choosing a game?**
   A. Risk level
   B. Type of risks
   C. User reviews
   D. Age recommendation
   E. Content explanations

9. **Which of the following is most difficult for you when evaluating a game**
   A. Understanding the actual level of risk
   B. Identifying hidden features
   C. Keeping up with new games
   D. Trusting official age ratings
   E. Nothing is difficult

10. **Would you prefer to receive alerts when a game's risk level increases**
   A. Yes
   B. No
   C. Not sure

Questionnaire Answers:

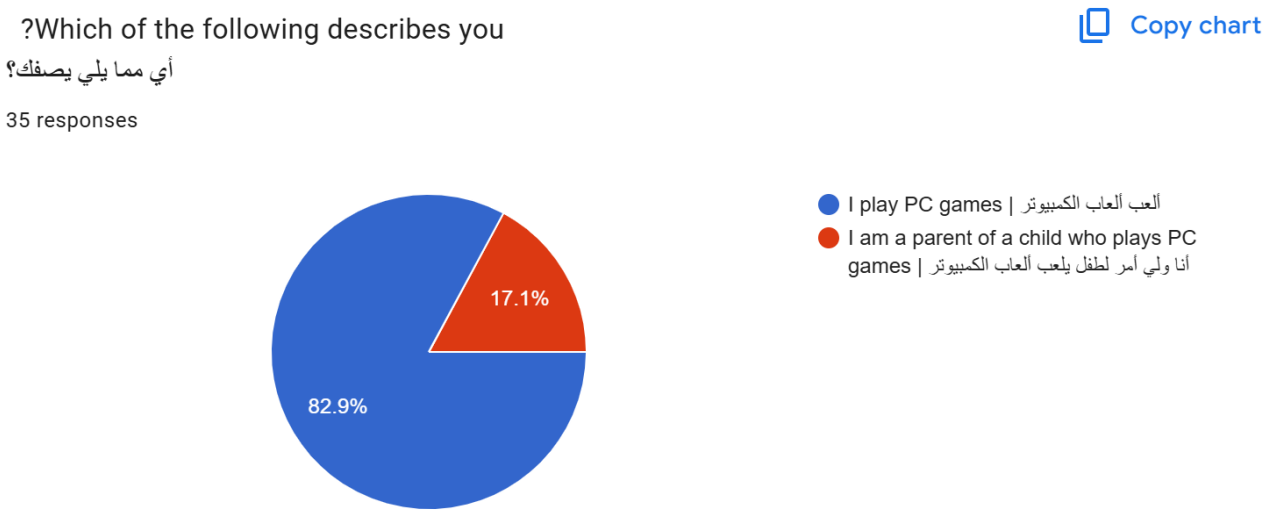

Figure 7-1: The pie chart of the first question

As Gamer:

لاعب | Gamer 🎮

? What is your age group
ما هي فئتك العمرية؟

29 responses

Copy chart

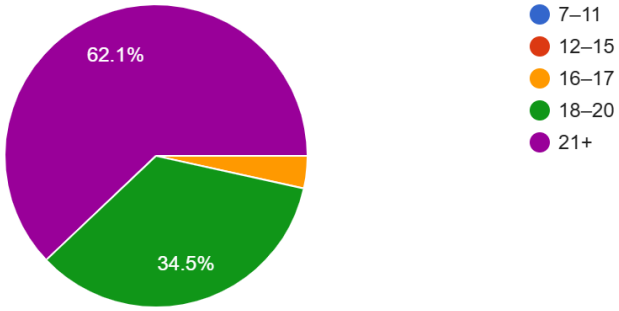

Figure 7-2: The pie chart of the second question

?How often do you play PC games
كم مرة تلعب ألعاب الكمبيوتر؟

29 responses

Copy chart

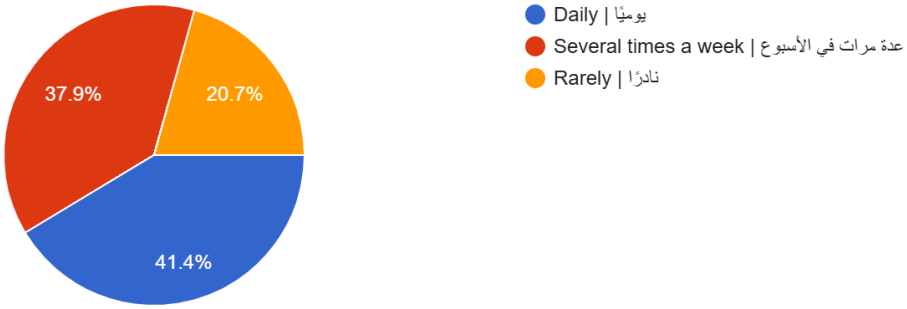

Figure 7-3:: The pie chart of the third question

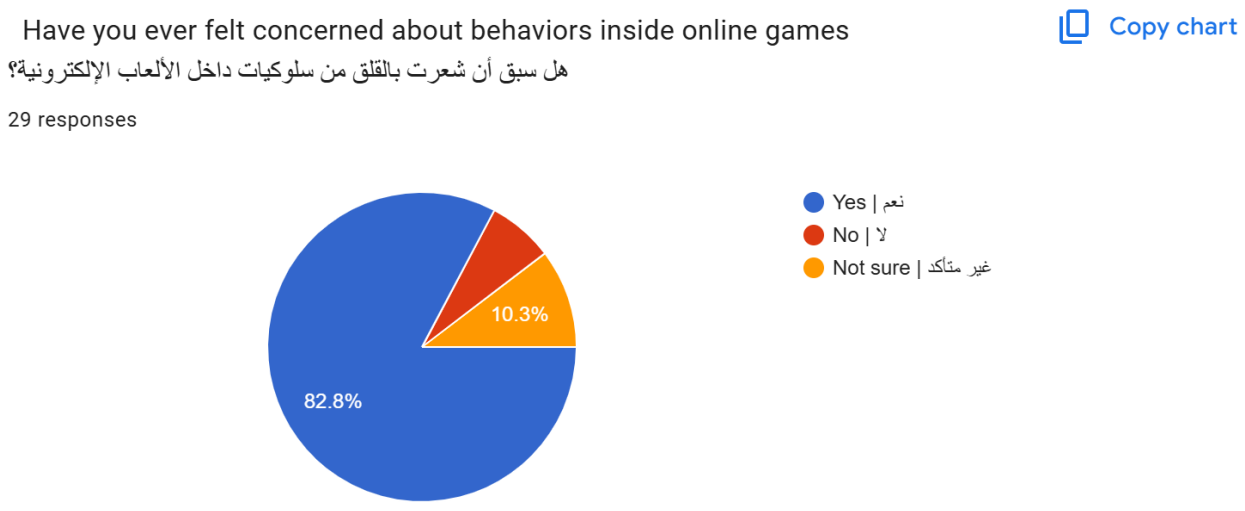

Figure 7-4: The pie chart of the fourth question

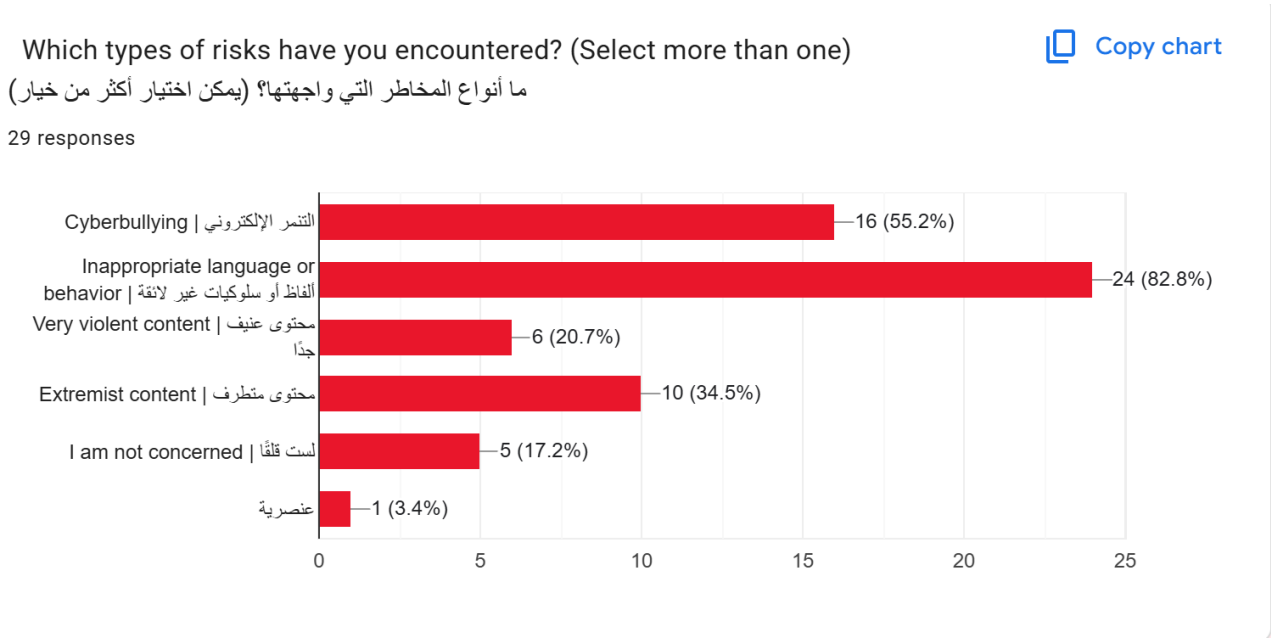

Figure 7-5: The diagram of the fifth question

Do you believe age ratings reflect the actual in-game environment
هل تعتقد أن تصنيفات العمر تعكس بيئة اللعبة الفعلية؟

Copy chart

29 responses

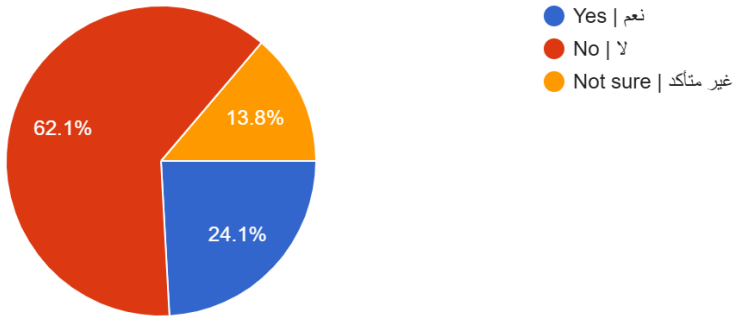

Figure 7-6: The pie chart of the sixth question

Would you like to know a game's risk level before playing it
هل تود معرفة مستوى خطورة اللعبة قبل لعبها؟

Copy chart

29 responses

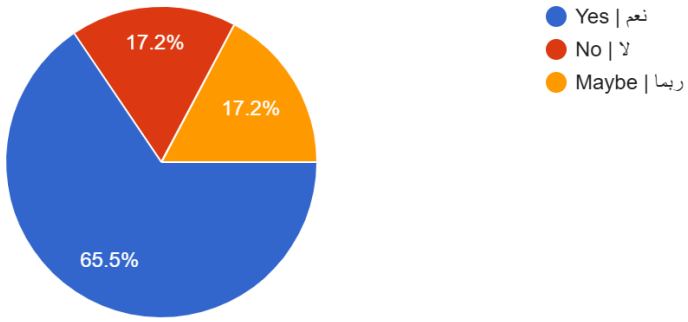

Figure 7-7The pie chart of the seventh question

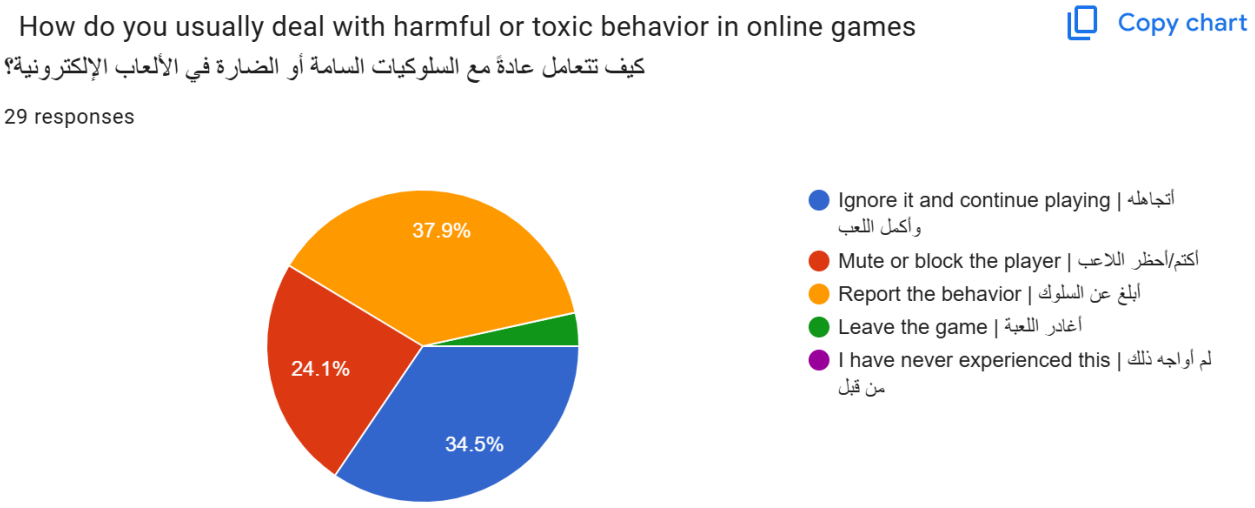

Figure 7-8:The pie chart of the eighth question

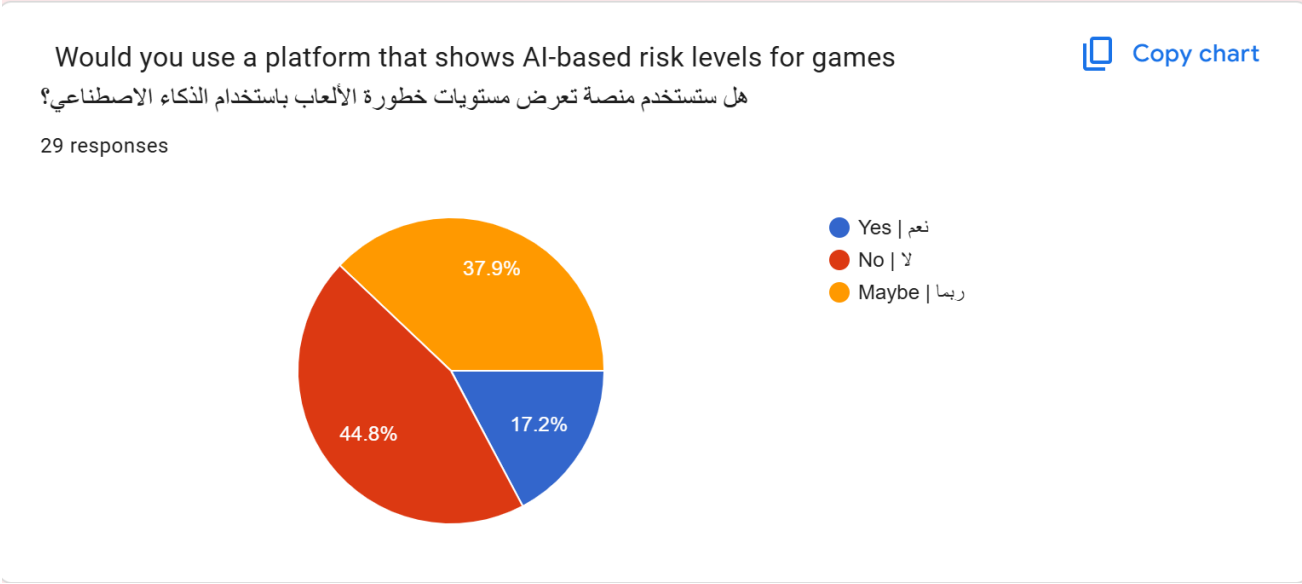

Figure 7-9The pie chart of the ninth question

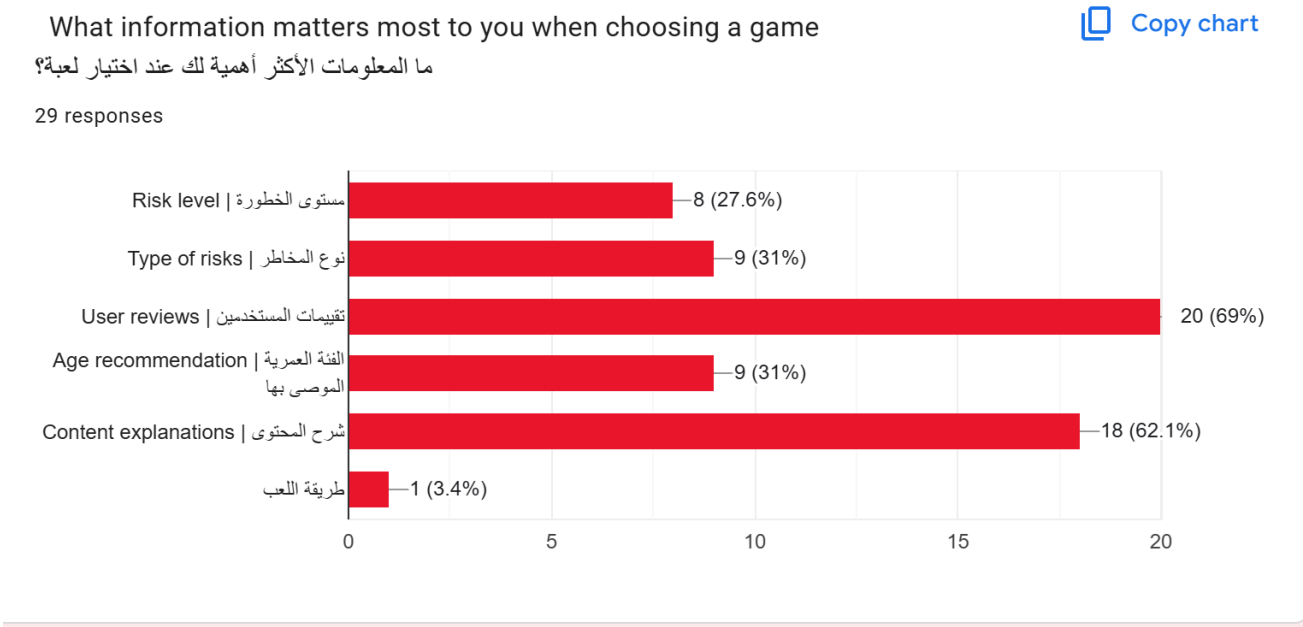

Figure 7-10: The diagram of the tenth question

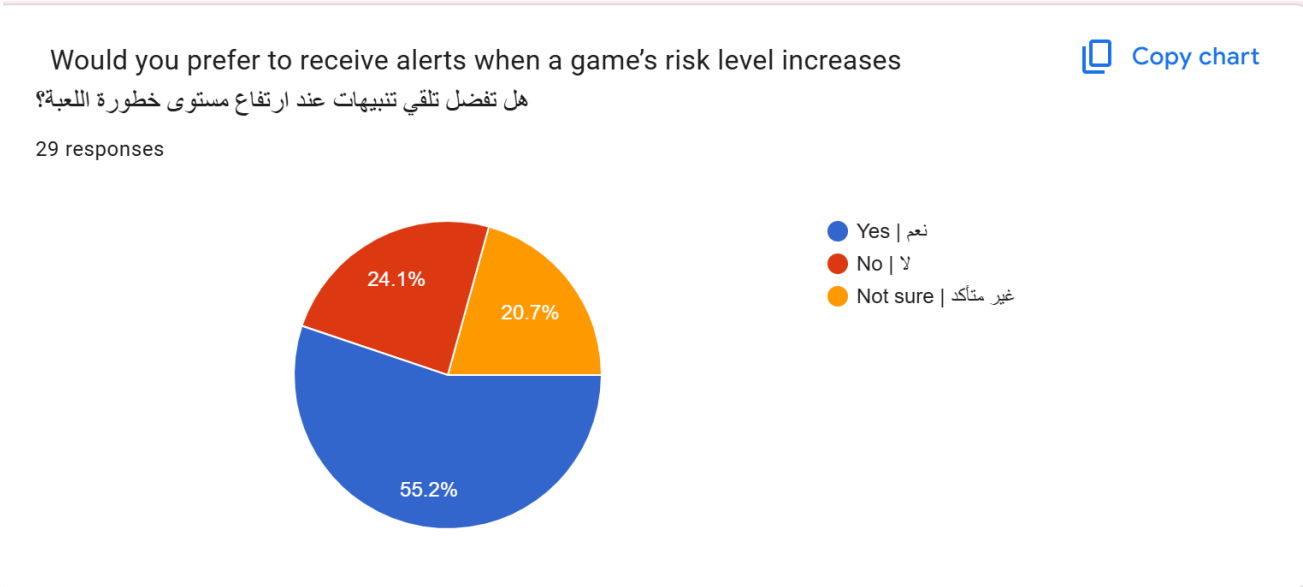

Figure 7-11: The pie chart of the eleventh question

As Parent:

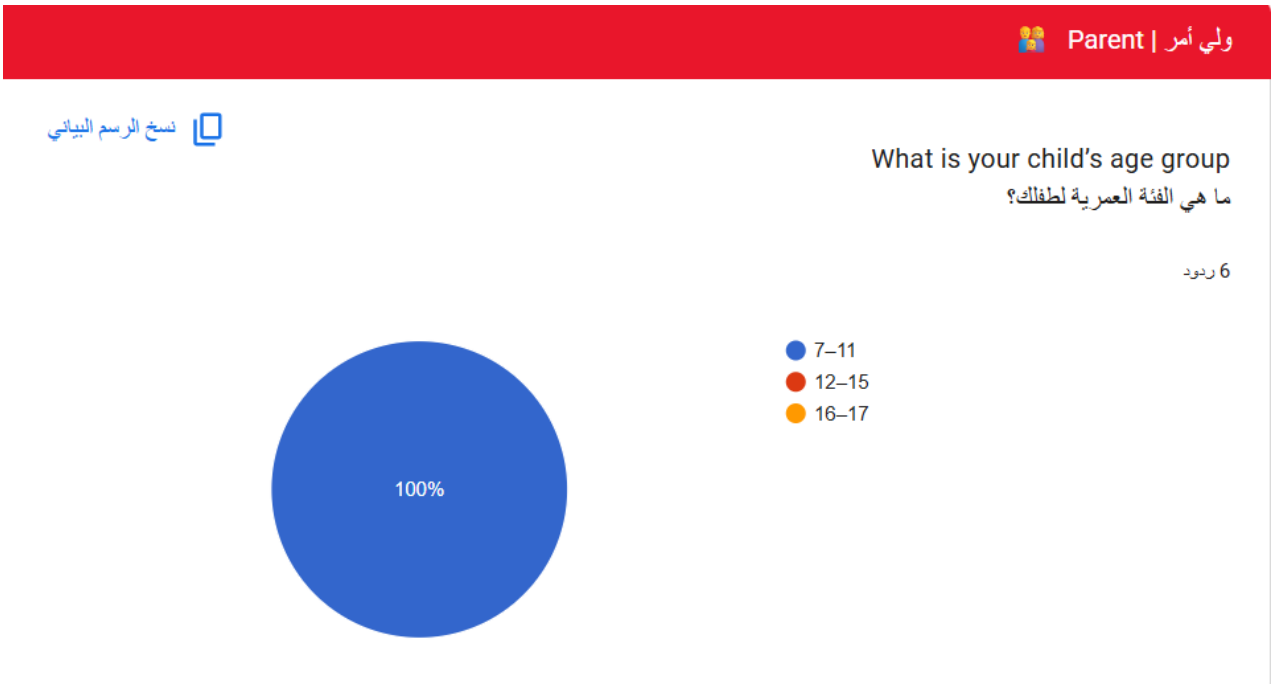

Figure 7-12: The pie chart of the twelfth question

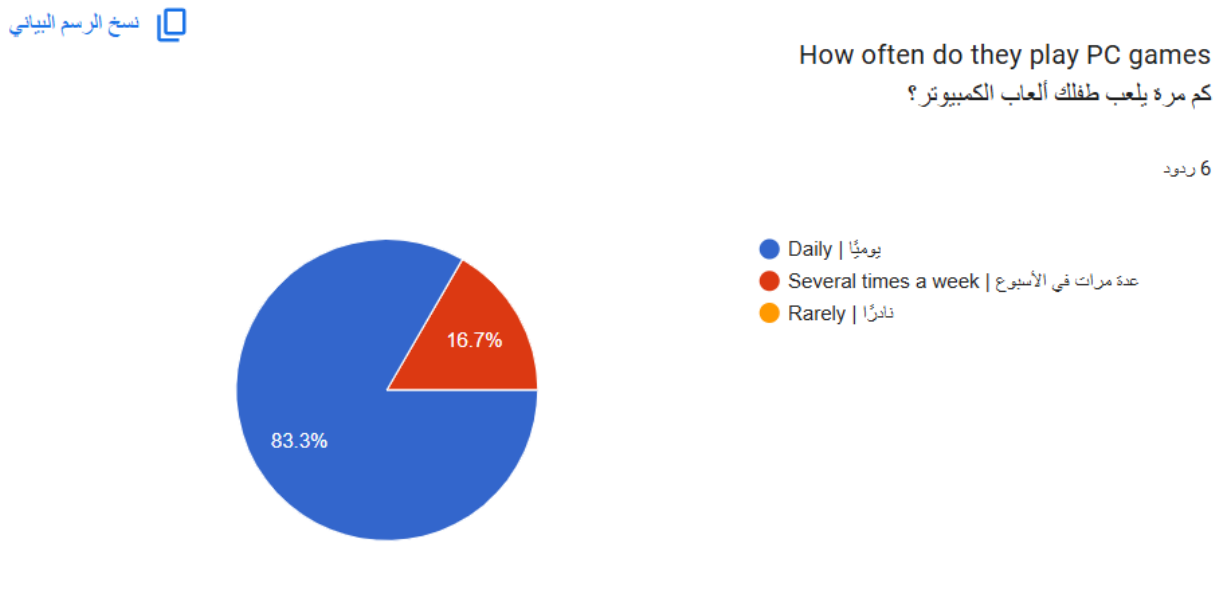

Figure 7-13: The pie chart of the thirteenth question

Figure 7-15: The pie chart of the fourteenth question

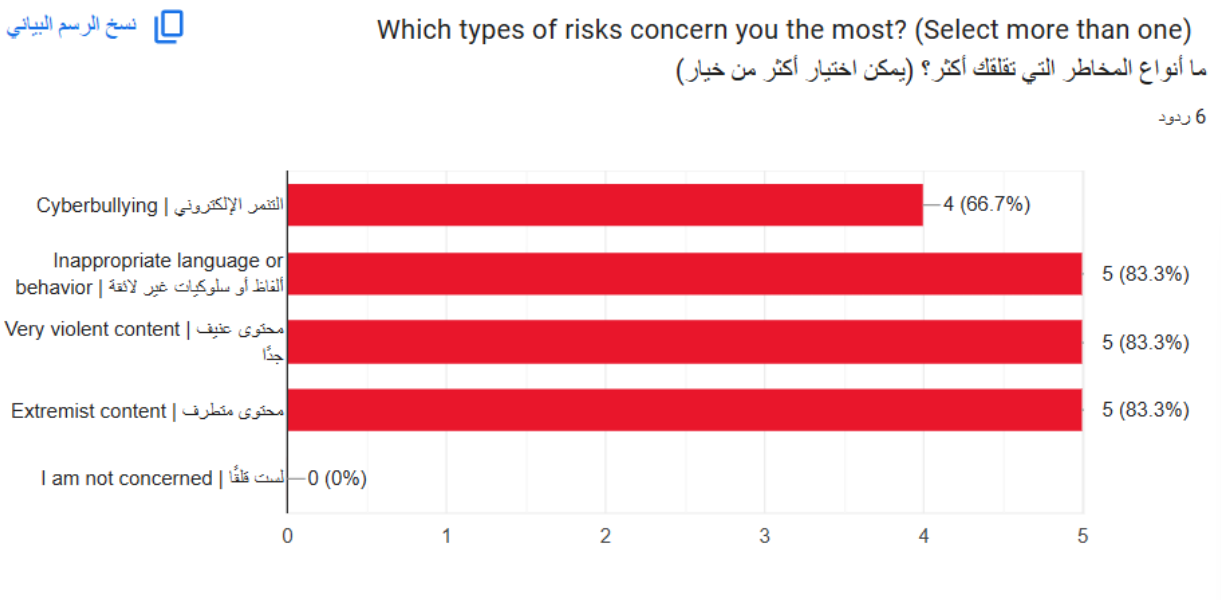

Figure 7-14: The pie chart of the fifteenth question

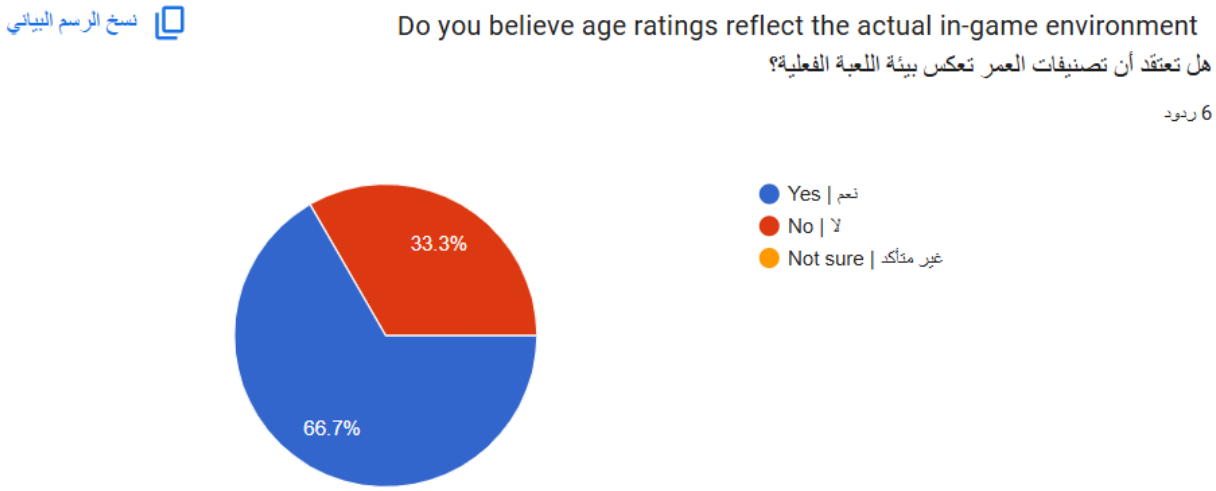

Figure 7-17: The pie chart of the sixteenth question

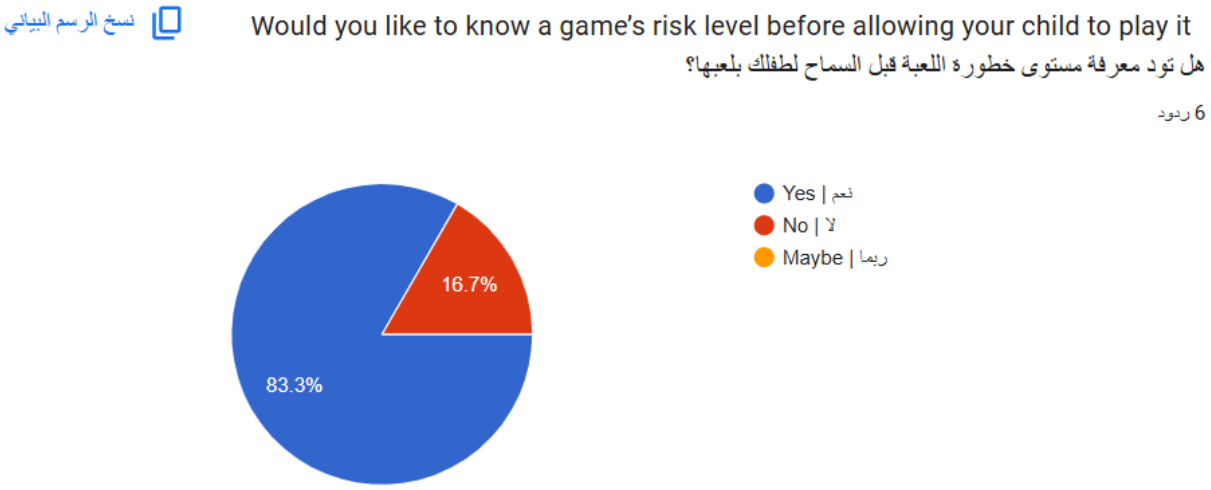

Figure 7-16: The pie chart of the seventeenth question

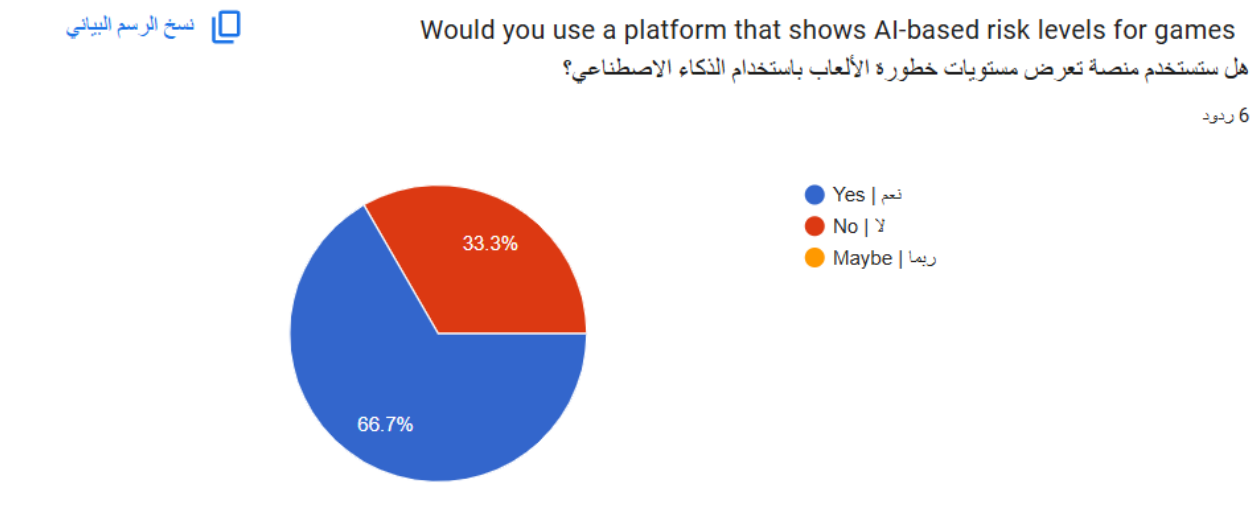

Figure 7-19: The pie chart of the eighteenth question

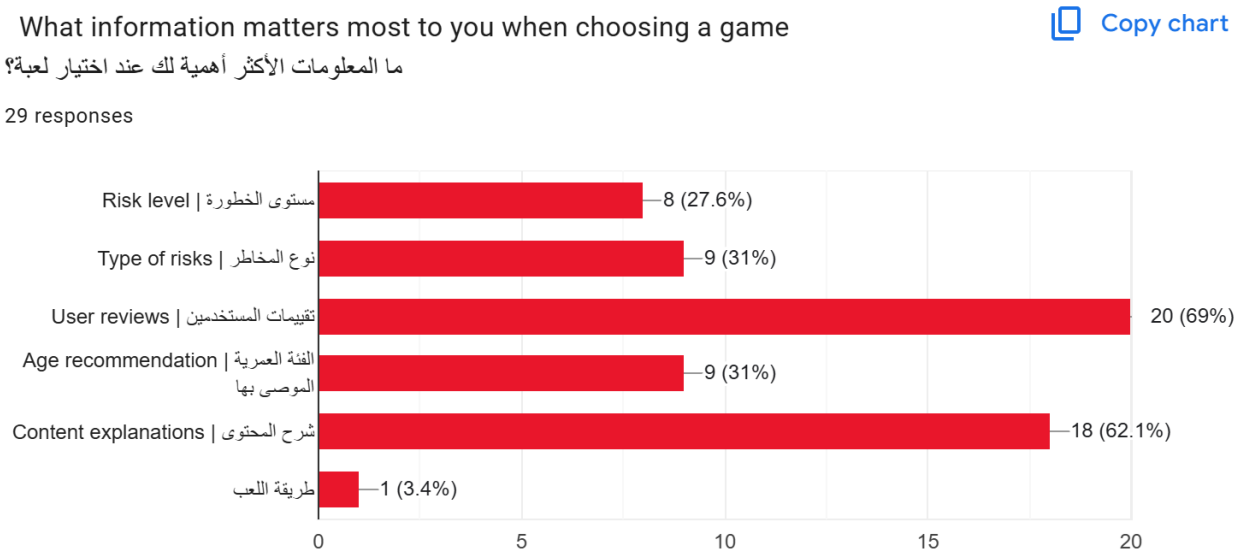

Figure 7-18: The pie chart of the ninetieth question

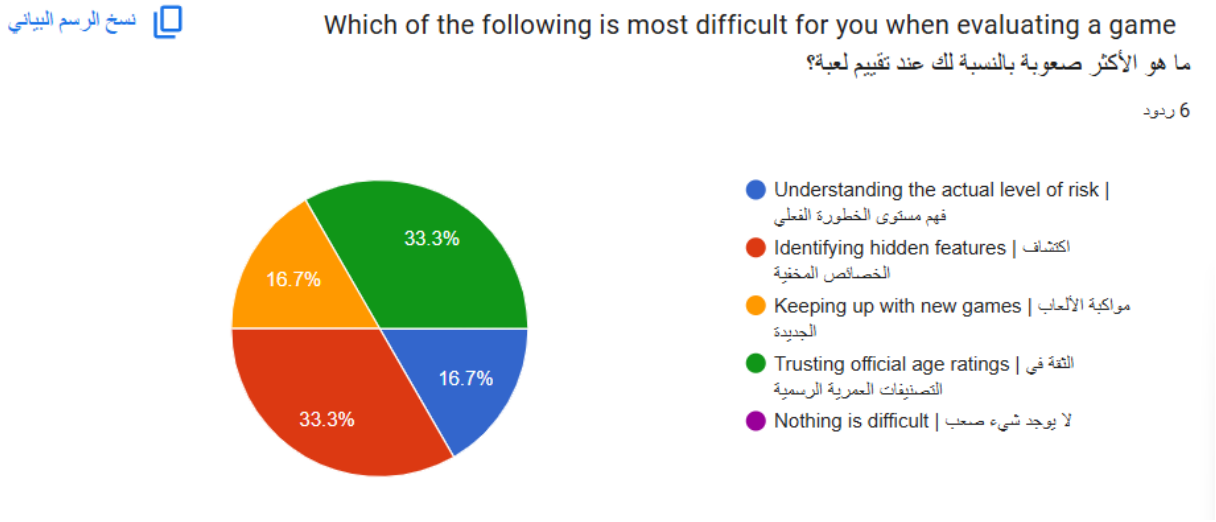

Figure 7-21: The pie chart of the twentieth question

Figure 7-20: The pie chart of the twenty-first question

7.3 APPENDIX C: Jira & GitHub [16][17]

This appendix demonstrates how we used Jira, a widely adopted project management tool, in our project workflow. Jira offers a powerful platform for organizing tasks, monitoring progress, and facilitating team collaboration. In this section, we will focus on the Product Backlog, which acts as a dynamic list of prioritized tasks that we plan to complete. And GitHub for adding the work.

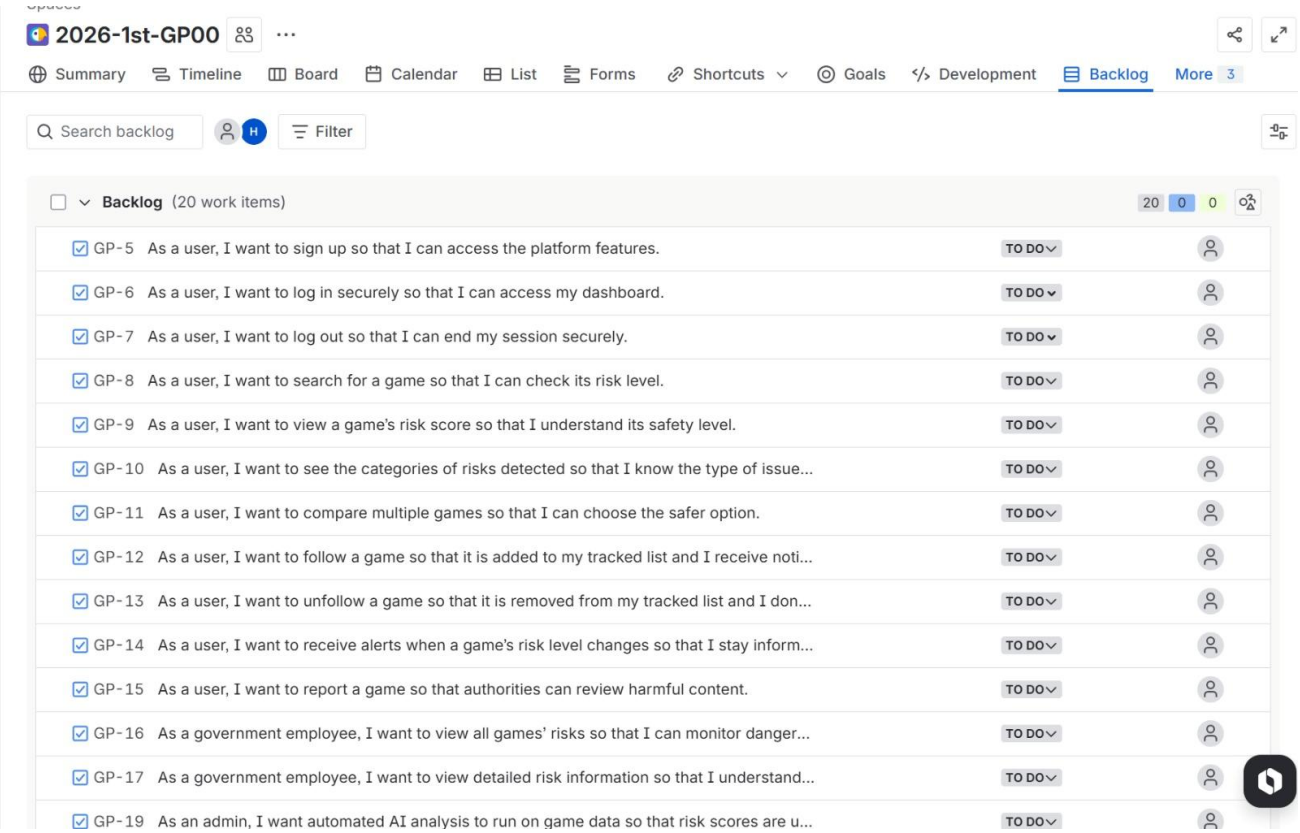

Figure 7-22: Backlog of Dir'a website in jira